

COMP5152

Advanced Data Analytics

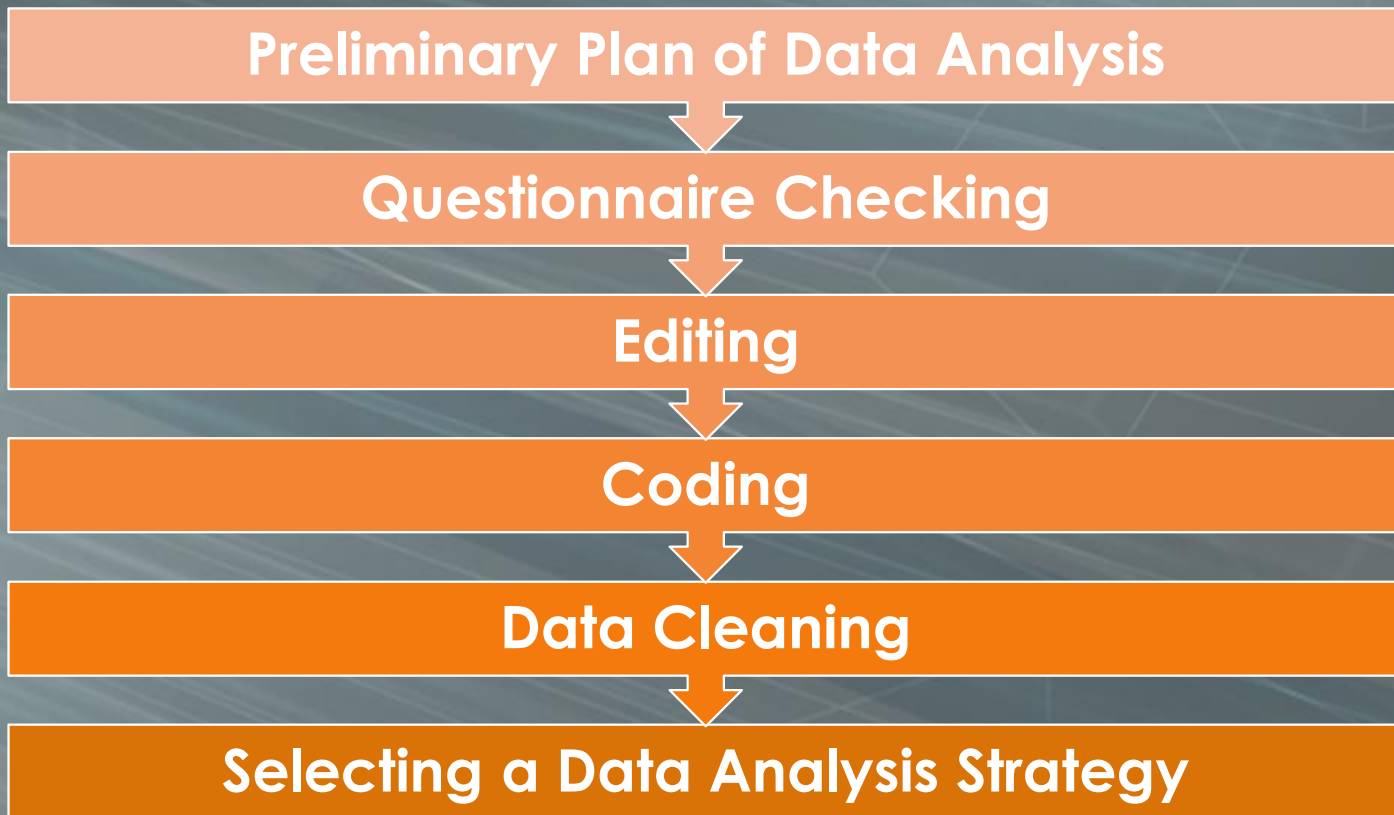
Types of Data and Statistic Basics

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Why Data and Statistics?

- Data provides the raw material for analysis.
- Statistics help in extracting meaningful insights from data
- Understand the nature of data in detail
- Understand some basics of statistics and then more advance in later lesson

Data Preparation Process : An Overview



Preliminary Plan

- Define Objectives & Stakeholders
 - ▶ Clarify project goals and identify who will use or be affected by the data.
- Assess Data Needs & Sources
 - ▶ Determine what data is required, where it can be obtained, and any access issues.
- Plan Sampling Strategy
 - ▶ Decide if sampling is needed, what sampling method to use, and how to ensure the sample represents the population.
- Resource & Timeline Planning
 - ▶ Estimate required resources, set a timeline, and anticipate potential challenges.

Questionnaire Checking

A questionnaire returned from the field may be **unacceptable** for several reasons:

- Parts of the questionnaire may be incomplete.
- The pattern of responses may indicate that the respondent did not understand or follow the instructions.
- The responses show little variance.
- One or more pages are missing.
- The questionnaire is received after the pre-established cut-off date.
- The questionnaire is answered by someone who does not qualify for participation.

Editing

Treatment of Unsatisfactory Results:

- Returning to the Field – The questionnaires with unsatisfactory responses may be returned to the field, where the interviewers recontact the respondents.
- Assigning Missing Values – If returning the questionnaires to the field is not feasible, the editor may assign missing values to unsatisfactory responses.
- Discarding Unsatisfactory Respondents – In this approach, the respondents with unsatisfactory responses are simply discarded.

Coding and Transcribing

- ▶ **Coding** means assigning a code, usually a number, to each possible response to each question.
 - ▶ The code includes an indication of the column position (field) and data record it will occupy.
 - ▶ Type of data do affect the coding (*more details in later section)

Example of Codebook

Column Number	Variable Number	Variable Name	Question Number	Coding Instructions
1	1	ID		1 to 20 as coded
2	2	Preference	1	Input the number circled. 1=Weak Preference 7=Strong Preference
3	3	Quality	2	Input the number circled. 1=Poor 7=Excellent
4	4	Quantity	3	Input the number circled. 1=Poor 7=Excellent
5	5	Value	4	Input the number circled. 1=Poor 7=Excellent
6	6	Service	5	Input the number circled. 1=Poor 7=Excellent

Example of Data

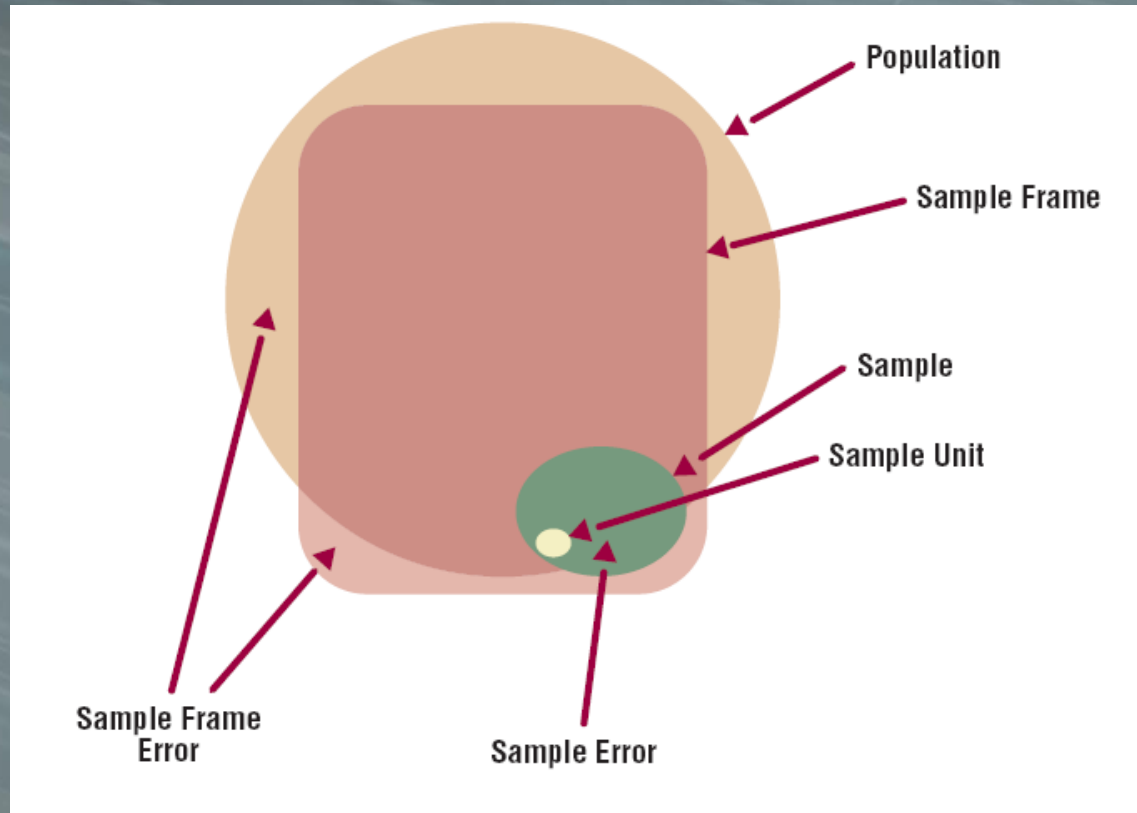
ID	PREFER.	QUALITY	QUANTITY	VALUE	SERVICE	INCOME
1	2	2	3	1	3	6
2	6	5	6	5	7	2
3	4	4	3	4	5	3
4	1	2	1	1	2	5
5	7	6	6	5	4	1
6	5	4	4	5	4	3
7	2	2	3	2	3	5
8	3	3	4	2	3	4
9	7	6	7	6	5	2
10	2	3	2	2	2	5
11	2	3	2	1	3	6
12	6	6	6	6	7	2
13	4	4	3	3	4	3
14	1	1	3	1	2	4
15	7	7	5	5	4	2
16	5	5	4	5	5	3
17	2	3	1	2	3	4
18	4	4	3	3	3	3
19	7	5	5	7	5	5
20	3	2	2	3	3	3

Data Cleaning & Statistically Adjusting the Data

- ▶ **Data Cleaning** - Thorough and extensive checks for consistency and treatment of missing responses or values.
- ▶ **Statistically Adjusting the Data** – The transformation of data to create new variables or the modification of existing variables so that they are more consistent with the objectives of the study and facilitate analysis, e.g. standardization and re-coding.

Data Collection and Sampling

Basic Concepts in Sampling



Basic Concepts in Sampling

- **Population:** the entire group under study as defined by research objectives
- A **census** requires information from everyone in the population.
- **Sample:** a subset of the population that should represent the entire group
- **Sample unit:** the basic level of investigation
- **A sample frame:** a master list of the entire population
- **Sampling frame error:** the degree to which the sample frame fails to account for all of the population
- **Why take sample not census?**

Types of Sampling

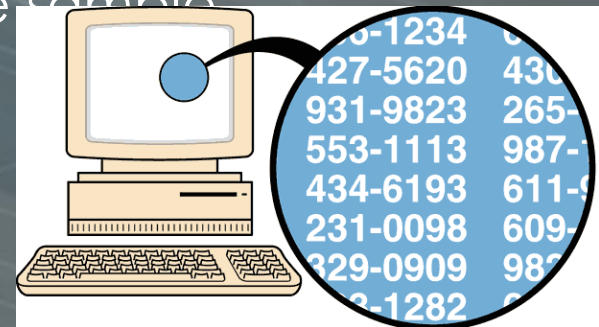
- ▶ **Probability Sampling**: Each unit in the target population has a **known probability** (usually equal chance) of being selected.
 - ▶ E.g. Simple random sampling
 - ▶ Adv: **Unbiased selection & Proper Sample Representation of target population.**
- ▶ **Nonprobability Sampling**: The probability of selecting each unit in target population is **unknown**.
 - ▶ E.g. convenience sampling, say intercept at high-end mall.
 - ▶ Adv: **Relatively quicker & lower cost**

Probability Sampling

Simple Random Sampling:

Easiest to understand method. Each unit of target population has an equal probability of being selected.

- Uses a table of random numbers, e.g. random digit dialling, SME business directory, customer list.
- Some other random selection procedure that guarantees that EACH member of the population has an identical chance of being selected into the sample.



Probability Sampling

Systematic Random Sampling:

- ▶ Using a LIST of the members of the population, the researcher selects a **random starting point** for the first sample member. A constant 'skip interval' is then used to select every other sample member.
- ▶ A skip interval = Target population size divide by sample size), e.g.
 - ▶ There are 100,000 elements in the population and a sample of 1,000 is desired. In this case the sampling interval, i , is 100. A random number between 1 and 100 is selected. If, for example, this number is 23, the sample consists of elements 23, 123, 223, 323, 423, 523, and so on.



Probability Sampling

Stratified Random Sampling:

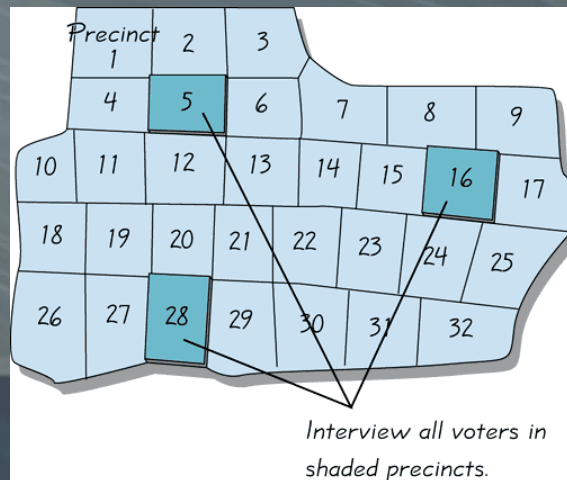
- We normally use it when we see clearly **homogeneous groups (strata)** for one distinguishing factor in the sampling frame (elements in each stratum are homogeneous; strata are different). As random sampling in EACH stratum, a representative sample will be surely obtained (number of sample in each stratum can be drawn according to population proportions, otherwise weighting can be applied on results).



Probability Sampling

Cluster Sampling:

- We usually use it when we see clearly a few **similar/identical clusters**, for each represents the population (elements in each cluster are heterogeneous). Hence, to save time and cost, you can indeed sample a few clusters first, then perform census or random sampling within each sampled cluster.
- E.g. **Area sampling** is one kind of cluster sampling.
- For example, to conduct exit interviews for Park'n Shop.



Probability Sampling

▶ **Cluster Sampling Example**

- ▶ You may assume all Park'n Shop in Central are having highly similar profile of customers. Then you can regard each Park'n Shop is a cluster. Then say there are 30 stores in central, you don't need to conduct exit interviews at every store in order to obtain a representative consumer feedback. You may select say 5 stores then do random sampling at each store.

Non-Probability Sampling

Convenience Sampling: respondents are selected because they happen to be in the right place at the right time.

- ▶ e.g. mall intercepts

Quota Sampling: quotas to be set to ensure a sufficient number of respondents to be obtained for a certain subgroup,

Characteristic	Population		Sample	
	Percentage		Percentage	Number
Male	48		48	480
Female	52		52	520
	100		100	1000

Non-Probability Sampling (cont')

Purposive Judgment Sampling: Sample is selected to fulfill a purpose based on researcher's/interviewer's judgment

- ▶ e.g. a test market conducted in a city that maybe regarded as representative of the country.

Referring or snowballing: Recruit by friend or respondents' referral, e.g. recruiting people who drink Macallan 18 years.

Strengths & Weaknesses of Basic Sampling Techniques

Technique	Strengths	Weaknesses
Nonprobability Sampling Techniques		
Convenience sampling	Least expensive; least time consuming; most convenient	Selection bias; sample not representative; not recommended for descriptive or causal research
Judgmental sampling	Low cost; convenient; not time consuming	No generalization; subjective
Quota sampling	Sample can be controlled for certain characteristics	Selection bias; No assurance of representativeness

Strengths & Weaknesses of Basic Sampling Techniques

Technique	Strengths	Weaknesses
Snowball sampling	Can estimate rare characteristics	Time consuming
Probability Sampling Simple random sampling (SRS)	Easily understood; results projectable	Difficult to construct sampling frame; expensive; lower Precision; no assurance of representativeness
Systematic sampling	Can increase representativeness; easier to implement than SRS; sampling frame not needed	Can decrease representativeness

Strengths & Weaknesses of Basic Sampling Techniques

Technique	Strengths	Weaknesses
Stratified sampling	Includes all Important Subpopulations; precision	Difficult to select relevant stratification variables; not feasible to stratify on many variables; expensive
Cluster sampling	Easy to implement; cost effective	Imprecise difficult to compute and interpret results

Choosing Nonprobability vs. Probability Sampling

Factors	CONDITIONS FAVORING THE USE OF	
	Nonprobability Sampling	Probability Sampling
Nature of research	Exploratory	Conclusive
Relative magnitude of sampling and nonsampling errors	Nonsampling errors are larger	Sampling errors are larger
Variability in the population	Homogeneous (low)	Heterogeneous (high)
Statistical considerations	Unfavorable	Favorable
Operational considerations	Favorable	Unfavorable

Factors for Determining Sample Design

- ▶ **Research objectives:** The type of research (exploratory, descriptive, or causal) and the **desire to be able to apply the results to a target population (need to generalize the findings)** will strongly influence the research design, including the sample design.
- ▶ **Degree of accuracy**
- ▶ **Resources (time & cost)**
- ▶ **Knowledge of the target population**
- ▶ **Scope of the research**
- ▶ **Statistical analysis needs**

Online Sampling Techniques

- ▶ **Online panels:** large numbers of individuals who have agreed to participate in online surveys
- ▶ **River samples:** created via the use of banners, pop-ups, or other online devices that invite website visitors to take part in the survey
- ▶ **E-mail list samples:** purchased or otherwise procured from someone or some company that has compiled email addresses of opt-in members of the population of interest

Develop a Sample Plan

- Define the population
- Obtain a sample frame
- Decide on the sample method
- Decide on the sample size
- Draw the sample
- Validate the sample

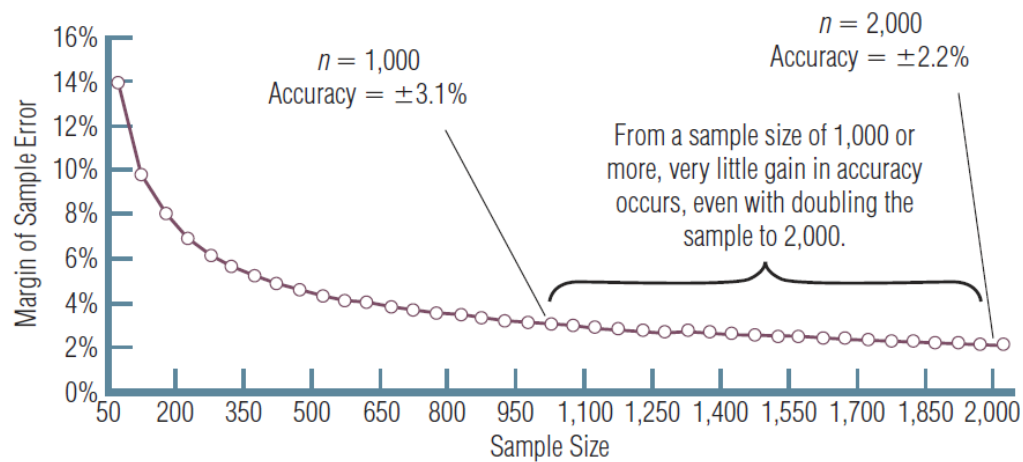
TABLE 10.1 The Axioms of Random Sample Size and Sample Accuracy

1. The only perfectly accurate sample is a census.
 2. A random sample will always have some inaccuracy, which is referred to as *margin of sample error* or simply *sample error*.
 3. The larger a random sample is, the more accurate it is, meaning the less margin of sample error it has.
 4. Margin of sample error can be calculated with a simple formula and expressed as a $\pm\%$ number.
 5. You can take any finding in the survey, replicate the survey with a random sample of the same size, and be “very likely” to find the same finding within the $\pm\%$ range of the original sample’s finding.
 6. In almost all cases, the margin of sample error of a random sample is independent of the size of the population.
 7. A random sample size can be a tiny percent of the population size and still have a small margin of sample error.
 8. The size of a random sample depends on the client’s desired accuracy (acceptable margin of sample error) balanced against the cost of data collection for that sample size.
-

Sample Size and Accuracy

- ▶ Which of these is more accurate?
 - ▶ A large probability sample or
 - ▶ A small probability sample?
- ▶ The larger a probability sample is, the more accurate it is (less sample error).

FIGURE 10.1
The Relationship
Between Sample Size
and Sample Error



How Population Size (N) Affects Sample Size

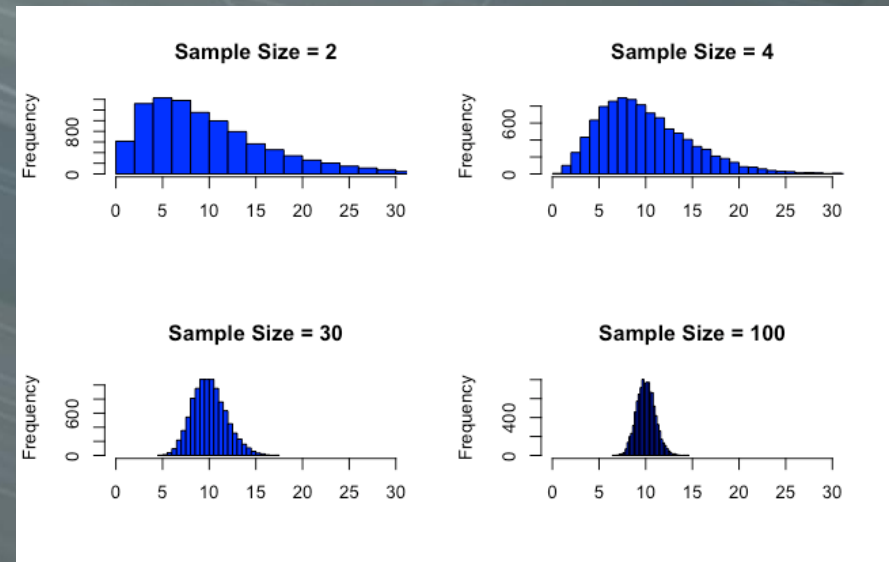
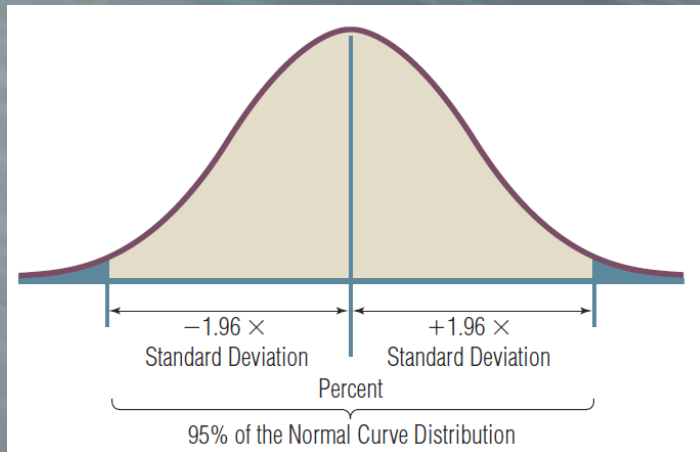
TABLE 10.2 Sample Sizes and Margin of Sample Error

Sample Size (<i>n</i>)	Margin of Sample Error (Accuracy Level)
10	±31.0%
50	±13.9%
100	±9.8%
200	±6.9%
400	±4.9%
500	±4.4%
750	±3.6%
1,000	±3.1%
1,500	±2.5%
2,000	±2.2%
5,000	±1.4%
10,000	±1.0%

Confidence Interval

Central Limit Theorem

- ▶ If we conducted our study over and over with a certain sample size x , we would expect our result to fall within a known range (the deviation). If we conduct the study with a larger sample size, this range will shrink, so the error is smaller.



Other Methods of Sample Size Determination

- ▶ Arbitrary "percent rule of thumb" sample size
 - ▶ Unscientific rule to decide sample size, such as "sample 10% of the population" or "take 100 samples regardless of population size."
- ▶ Conventional sample size specification
 - ▶ Based on field/industry norms
- ▶ Statistical analysis requirements sample size specification
 - ▶ certain statistical tests require a minimum number of observations to be valid or to achieve a desired level of power or precision.
- ▶ Cost basis of sample size specification
 - ▶ Sample size is determined by the available budget or resources.

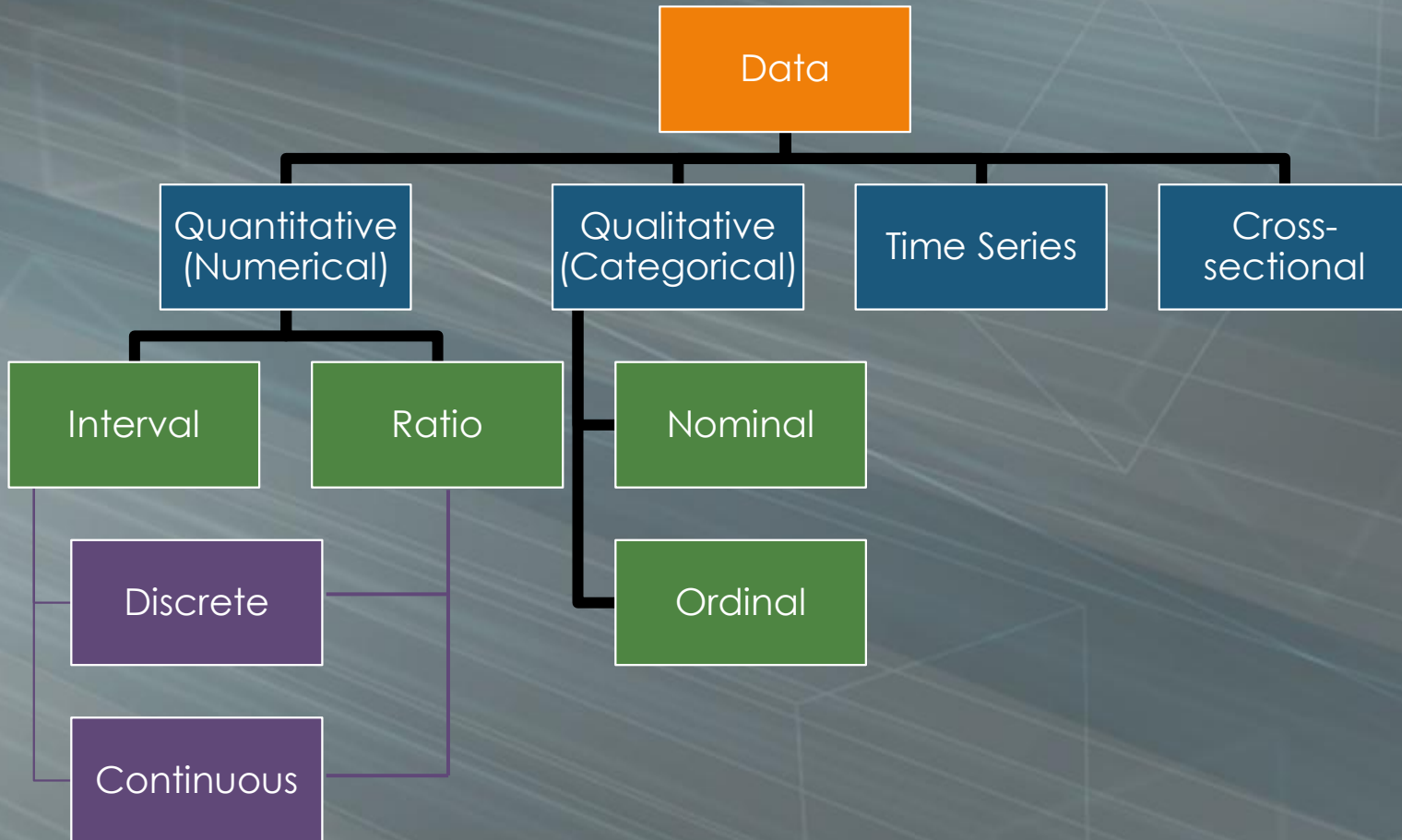
Types of Data and Data Coding

Types of Data

In the realm of statistics we have only four, known as the 'scales of measurement'

- ▶ Nominal
- ▶ Ordinal
- ▶ Interval
- ▶ Ratio

Types of Data



Types of Data

▶ **Qualitative** data

- ▶ Observations that are non-numerical or categorical
- ▶ Either the nominal or ordinal scale of measurement

▶ **Quantitative** data

- ▶ Observations that are numerical, either discrete or continuous
- ▶ Interval & Ratio scale of measurement

Quantitative

Age - income

55	75000
42	68000
.	.
.	.

Number of defects

0
3
12
.

Qualitative

Person Married/unmarried

Tom	yes
John	no
Sam	no
.	.
.	.

Student Grade

1	A
2	B+
3	D
.	.
.	.

Quantitative Data

- ▶ **Discrete** quantitative data
 - ▶ Data usually result from counting
 - ▶ E.g., number of customers arrived in 2 mins
- ▶ **Continuous** quantitative data
 - ▶ Data usually result from measuring
 - ▶ E.g., elapsed time of a production run, your height

Both type of data can be used for **arithmetic calculation**

Qualitative Data

▶ **Nominal**

- ▶ Only for classification of observations and no particular order would be identified
- ▶ E.g., colour (red, green, blue, and yellow)
- ▶ cannot be used for arithmetic calculation

▶ **Ordinal** (Ranked)

- ▶ For classification of observations
- ▶ Can be used for arithmetic calculation by its ordering process
- ▶ E.g., strongly agree, agree, neutral, disagree, strongly disagree (Likert scale)

Quantitative Data

▶ Interval

- ▶ like the ordinal level, with the additional property that the difference between any two data values is meaningful, however, there is no natural zero starting point (where none of the quantity is present)
- ▶ E.g. : Years 1000, 2000, 1776, and 1492

Quantitative Data

▶ Ratio

- ▶ the interval level with the additional property that there is also a natural zero starting point (where zero indicates that none of the quantity is present); for values at this level, differences and ratios are meaningful
- ▶ E.g. : Prices of college textbooks (\$0 represents no cost, a \$100 book costs twice as much as a \$50 book)

Types of Data

▶ **Time Series** Data

- ▶ Data values that are recorded in a 'regular' successive interval of time.
- ▶ Example: Tracking the closing prices of a particular stock over a month.

▶ **Cross-sectional** Data

- ▶ Data values that are measured at the same or approximately the same point of time.
- ▶ Example: The closing prices of 20 randomly selected stocks on a particular trading day.

Coding

- ▶ Coding – process of translating data from questionnaires or other sources into other representation easy for further analysis
 - ▶ E.g. for feed into machine learning models
- ▶ Label which is not a numerical
- ▶ Coding can make data more consistent:
 - ▶ Example: Question = Sex
 - ▶ Answers = Male, Female, M, or F
 - ▶ Coding will avoid such inconsistencies

Coding Systems

- ▶ Common coding systems (code and label) for dichotomous variables:
 - ▶ 0=No 1=Yes
(1 = value assigned, Yes= label of value)
 - ▶ OR: 1=No 2=Yes
- ▶ When you assign a value you must also make it clear what that value means
 - ▶ In first example above, 1=Yes but in second example 1=No
 - ▶ As long as it is clear how the data are coded, either is fine
- ▶ You can make it clear by creating a data dictionary to accompany the dataset

Coding:

Attaching Labels to Values

- ▶ Many analysis software packages allow you to attach a label to the variable values
Example: Label 0's as male and 1's as female
- ▶ Makes reading data output easier:

Without label:	Variable SEX	Frequency	Percent
	0	21	60%
	1	14	40%

With label:	Variable SEX	Frequency	Percent
	Male	21	60%
	Female	14	40%

Coding- Ordinal Variables

- ▶ Coding process is similar with other categorical variables
- ▶ Example: variable EDUCATION, possible coding:
 - 0 = Primary school
 - 1 = Secondary school
 - 2 = University
 - 3 = Graduate school
- ▶ Could be coded in reverse order (0=Graduate school, 3=Primary school)
- ▶ For this ordinal variable we want numbering somehow reflect the education level

Coding – Ordinal Variables (cont.)

- ▶ Example of bad coding:
 - 0 = University
 - 1 = Graduate school
 - 2 = Secondary school
 - 3 = Primary school
- ▶ Data has an inherent order but coding does not follow that order—NOT appropriate coding for an ordinal categorical variable

Coding: Nominal Variables

- ▶ For coding nominal variables, order makes no difference
- ▶ Example: variable FRUIT
 - 1 = Apple
 - 2 = Orange
 - 3 = Mango
 - 4 = Banana
- ▶ Order does not matter, no ordered value associated with each response

Coding: Continuous Variables

- ▶ Creating categories from a continuous variable (ex. age) is common
- ▶ May break down a continuous variable into chosen categories by creating an ordinal categorical variable
- ▶ Example: variable = AGECAT
 - 1 = 0–9 years old
 - 2 = 10–19 years old
 - 3 = 20–39 years old
 - 4 = 40–59 years old
 - 5 = 60 years or older

Label Encoding for Machine Learning

- ▶ It is important to convert categorical data (i.e. Label) into numerical ones
- ▶ So as to fit to ML models which only take numerical data
- ▶ Take the previous EDUCATION variable as example, it is an Ordinal Encoding
 - 0 = Primary school
 - 1 = Secondary school
 - 2 = University
 - 3 = Graduate school

One-hot Encoding

- ▶ But Ordinal Encoding may **NOT** be good if your categorical data is a nominal one
 - ▶ As the numerical values 0-3 has order, which may confuse ML models
- ▶ One hot encoding is often used if Nominal data is to be encoded

Fruit	One-Hot Encoding			
Apple	1	0	0	0
Orange	0	1	0	0
Mango	0	0	1	0
Banana	0	0	0	1

Advantages of One-hot Encoding

- ▶ It can improve model performance by providing more information to the model about the categorical variable
- ▶ It can help to avoid the problem of ordinality, not to confuse the ML models

Disadvantage of One-hot Encoding

- ▶ It can lead to increased dimensionality, as a separate column is created for each category in the variable.
 - ▶ This can make the model more complex and slow to train and easily overfit if sample size is small.
- ▶ It can lead to sparse data and large data size, as the encoding has large number of 0

Statistics

What is Statistics?

- ▶ **Statistics** is the study of the collection, organization, presentation and interpretation (analysis) of data
- ▶ Easy to describe an individual's feature, like his age, height, walking speed and etc.
- ▶ But how about a group of people?

Key Concepts and Terms in Statistics

1. Population (Universe)

- ▶ All items of interest

2. Sample

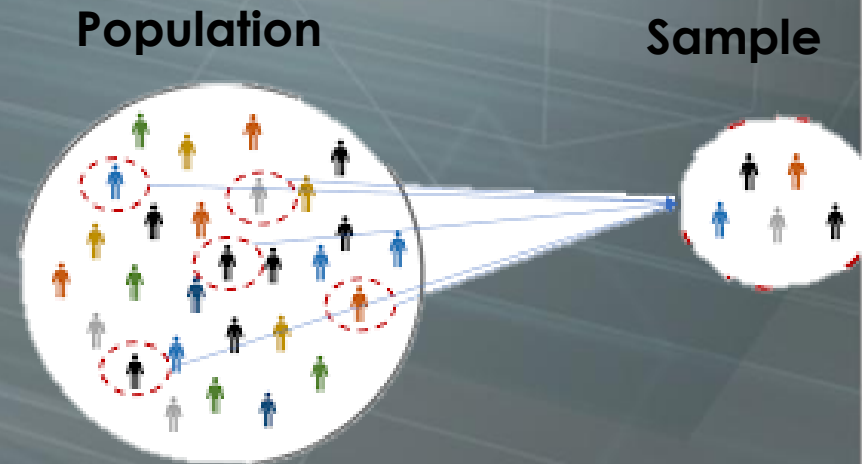
- ▶ A portion of a population

3. Parameter

- ▶ Summary measures of a population

4. Statistics

- ▶ Summary measures of a sample



Population and Sample

▶ **Population**

- ▶ The set of all items of interest involved in a survey
- ▶ A descriptive measure about the characteristic of a population is called a **parameter**

▶ **Sample**

- ▶ A set of data drawn from the population in order to estimate the parameter
- ▶ A descriptive measure of the sample is called a **statistics**

Example

- ▶ The manager of a smart phone manufacturer wishes to find out what proportion of smart phone owners is using their newly-released product, the Green Phone.
- ▶ He has randomly selected 500 smart phone owners and found that 175 of them are using Green Phone.

Population	all smart phone owners
Parameter	the proportion of all smart phone owners that are using green phone
Sample	the 500 smart phone owners that are surveyed
Statistics	the proportion of the 500 smart phone owners that are using green phone

Example (cont')

- ▶ Based on the survey results, if the manager reported to the senior management that the market share of green phone is about 35%.
- ▶ He has actually generalized the information (statistics) collected from a smaller group (sample) and judges that it (parameter) is the same for the larger designated group (population).
- ▶ This is **statistical inference (inferential statistics)**.

Inference: the act or process of reaching a judgement of whether something is true from known facts or evidence

Activity

The production manager of a manufacturer producing badminton shuttlecocks claims that the average weight of the shuttlecock produced is 5 grams. In five randomly selected tubes (a total of 60), the average weight of the shuttlecocks is found to be 4.83 grams.

- a) Identify the population and the sample.
- b) Is the value 5 grams corresponds to a parameter or a statistic?
- c) Is the value 4.83 grams corresponds to a parameter or a statistic?

Descriptive & Inferential Statistics

▶ **Descriptive Statistics**

- ▶ Concerns with summarizing and describing information from data collected
- ▶ Relates to data organization and display

▶ **Inferential Statistics**

- ▶ Concern with estimating, predicting and making inference about the parameter of a population base on information reflect from the statistics of a sample. (a process of inference)
- ▶ Relates to generalization of data in a wider scope

Overview of Statistical Methods



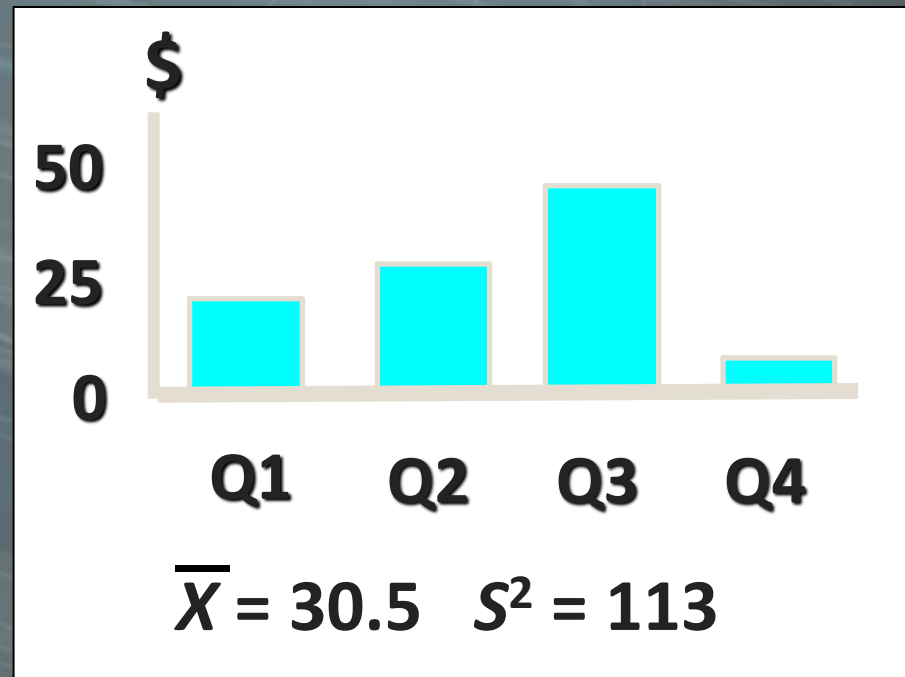
Descriptive Statistics

1. Involves

- ▶ Collecting data
- ▶ Summarizing data
- ▶ Presenting data

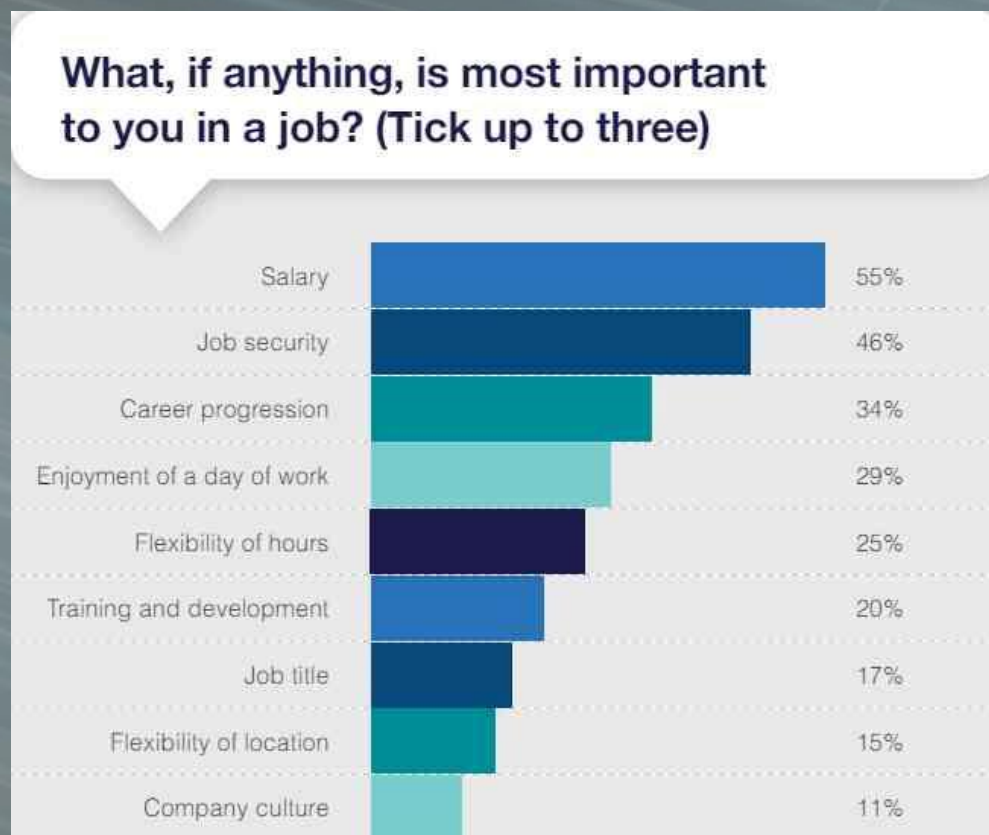
2. Purpose

- ▶ To organise data and try to dig out useful but often hidden information



Descriptive Statistics

- ▶ ADP surveyed 34,612 employees in 18 countries around the world from October 22 to November 24, 2023, with the Asia-Pacific region covering Australia, China, India, Japan and Singapore.



Inferential Statistics

1. Involves

- ▶ Estimation
- ▶ Hypothesis Testing

2. Purpose

- ▶ To get to understand the concerned characteristic of a population that is represented by a specific value (parameter) by analyzing similar value (statistics) from the sample collected.

Inferential Statistics

Michael Leiter shows weekly meetings and open employee-supervisor discussions reduced gossip, improved job satisfaction and reduced absenteeism 38 per cent in a health care setting.

Source:

Leiter, Michael P., Heather K. Spence Laschinger, Arla Day, and Debra Gilin Oore. (2011). The Impact of Civility Interventions on Employee Social Behavior, Distress, and Attitudes. *Journal of Applied Psychology* 96(6): 1258-1274.x

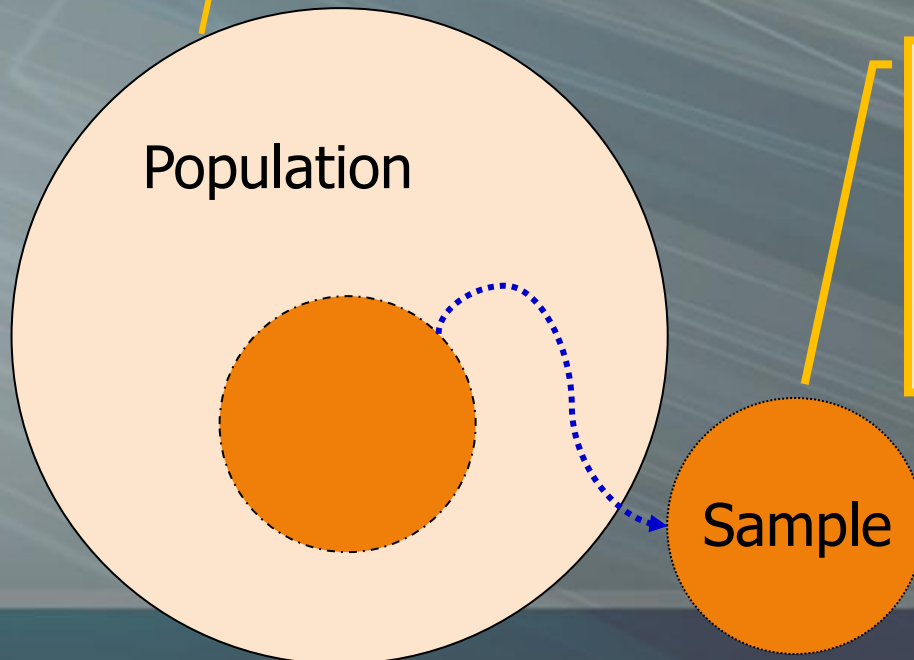
Jinseok Chun and colleagues find employees who believe their organization is fair and just are more loyal to the company and more willing to help fellow employees.

Source:

Chun, Jinseok, Yuhjung Shin, Jin Nam Choi, and Min Soo Kim. (2011) How Does Corporate Ethics Contribute to Firm Performance? The Mediating Role of Collective Organizational Commitment and Organizational Citizenship Behavior. *Journal of Management*, 20, 1-25.

Statistical Inference

Suppose that we are interested to find out the population mean value μ (the population mean, a **parameter**).

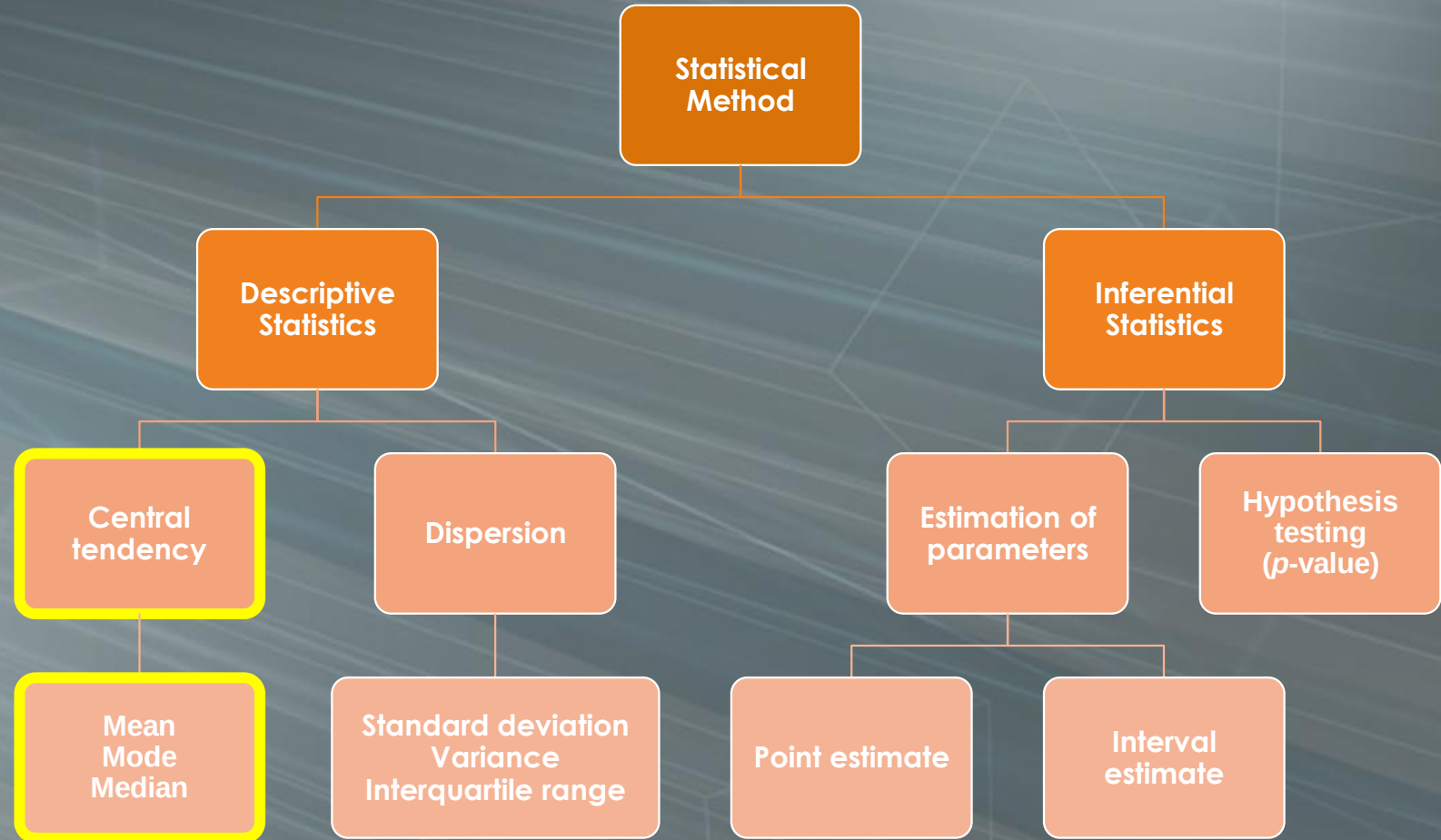


To save time and effort, we 'select' a sample from the population and obtain the sample mean $\bar{X}$ (a **statistic**).

Statistical Inference

- ▶ As long as the selection of the sample units is performed under an appropriate randomization process, the sample obtained is considered to be representative of the population. The sample mean $\bar{X}$ will then be a good estimate to the population mean μ . We will then make use of $\bar{X}$ to 'guess' or 'infer' what μ is about.
- ▶ The whole process used to draw conclusions about the characteristic (a parameter) of the population based on relevant sampled information (statistics) is called Statistical Inference.

Overview of Statistical Methods



Measures of Central Tendency

- ▶ What do they tell us?
 - ▶ To describe/summarise your data
 - ▶ Best represent your data
- ▶ We can use
 - ▶ Arithmetic mean* (average value)
 - ▶ Mode (most common value)
 - ▶ Median (middle value)



* geometric mean; harmonic mean; trimmed mean

Mean or Average

- ▶ Mean is the most popular and useful measure of central location.

$$\text{Mean} = \frac{\text{Sum of the measurements}}{\text{Number of measurements}}$$

Sample mean

$$\bar{X} = \frac{\sum_{i=1}^n x_i}{n}$$

Sample size, n

Population mean

$$\mu = \frac{\sum_{i=1}^N x_i}{N}$$

Population size, N

Mean or Average

- ▶ The mean of the sample of six measurements 7, 3, 9, -2, 4, 6 is given by

$$\bar{x} = \frac{\sum_{i=1}^6 x_i}{6} = \frac{7 + 3 + 9 - 2 + 4 + 6}{6} = 4.5$$

- ▶ Problems
 - ▶ With small samples a single outlier value can skew the average

Median or middle value

- ▶ The middle value but....
 - ▶ Sort your data numerically from smallest to largest
 - ▶ 19, 20, 18, 18, 58 becomes 18, 18, 19, 20, 58
 - ▶ So 19 is the middle or median value
 - ▶ 19 is more representative than 26.6 for the majority of people in our data set
 - ▶ If we have an even number of data, get the average of the middle two values.

Median

- ▶ The median of a set of measurements is the value that falls in the middle when the measurements are arranged in order of magnitude. For example, seven employee salaries were recorded as (in unit of \$1k):

28, 100, 26, 32, 30, 26, 29.

Find the median salary.

First, sort the salaries. Then, locate the middle value:

26, 26, 28, 29, 30, 32, 100

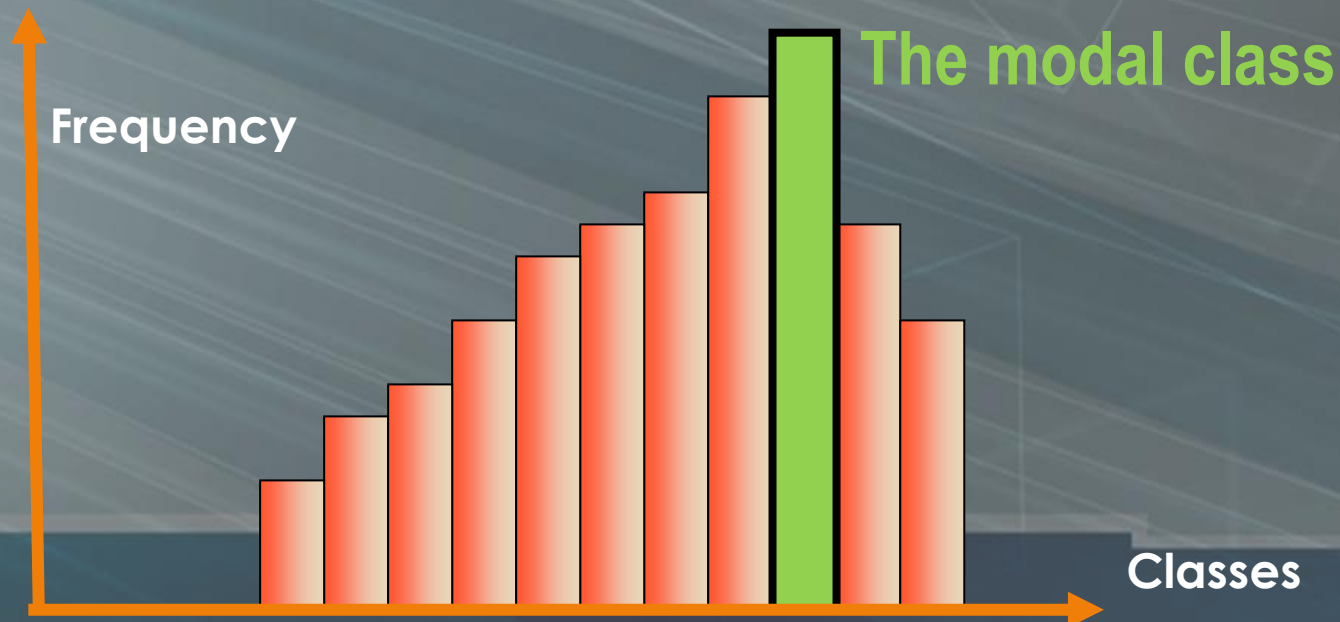
- ▶ Median = 29

Median (cont')

- ▶ Suppose one employee's salary of \$31,000 is added to the group recorded before. Find the median salary.
- ▶ Again, sort the salaries. Since the number of observations is **even**, there are two middle values, 29 and 30. The median is given by the average of these two middle values.
26, 26, 28, **29, 30**, 31, 32, 100
- ▶ Median = $(29 + 30) / 2 = 29.5$
- ▶ Median is not affected by any extreme observations but is only affected by the number of observations. How about the Mean?

Mode

- ▶ The mode of a set of measurements is the value that occurs most frequently
- ▶ E.g. Mode of the following dataset:
31, 34, 36, 33, 28, 34, 30, 34, 32, 40.
 - ▶ which is 34 (the modal class)
- ▶ Less used than mean or median



Comparisons

- ▶ **Mode** is especially useful in describing nominal and ordinal levels of measurement. If marks are classified by letter grade, the frequency of each grade can be counted. Then, the mode becomes a logical measure to compute.
- ▶ **Median** indicates that half of the class received a grade below 81%, and half of the class received a grade above 81%.
- ▶ **Mean** provides information about the over-all performance level of the class. It can serve as a tool for making comparisons with other classes and/or other exams.

Distribution

- ▶ When working with large data sets, it is often helpful to organize and summarize data by constructing a table/histogram called a frequency distribution.
- ▶ Frequency Distribution shows how a data set is partitioned among all categories (or classes) by listing all of the categories along with the number of data values in each of the categories.

Example

Pulse Rates of Females and Males

Raw/ Original Data

Table 2-1 Pulse Rates (beats per minute) of Females and Males

Females

76	72	88	60	72	68	80	64	68	68	80	76	68	72	96	72	68	72	64	80
64	80	76	76	76	80	104	88	60	76	72	72	88	80	60	72	88	88	124	64

Males

68	64	88	72	64	72	60	88	76	60	96	72	56	64	60	64	84	76	84	88
72	56	68	64	60	68	60	60	56	84	72	84	88	56	64	56	56	60	64	72

Example (cont')

Frequency Distribution Pulse Rates of Females

Table 2-2 Pulse Rates of Females

Pulse Rate	Frequency
60-69	12
70-79	14
80-89	11
90-99	1
100-109	1
110-119	0
120-129	1

The *frequency* for a particular **class** is the number of original values that fall into that class.

Lower Class Limits

are the smallest numbers that can actually belong to different classes

Lower Class
Limits

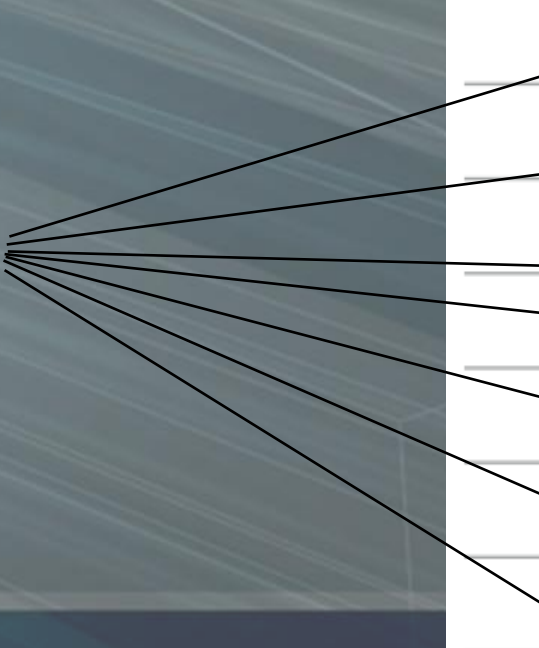


Table 2-2 Pulse Rates of Females

Pulse Rate	Frequency
60-69	12
70-79	14
80-89	11
90-99	1
100-109	1
110-119	0
120-129	1

Upper Class Limits

are the largest numbers that can actually belong to different classes

Upper Class
Limits

Table 2-2 Pulse Rates of Females

Pulse Rate	Frequency
60-69	12
70-79	14
80-89	11
90-99	1
100-109	1
110-119	0
120-129	1

Class Midpoints are the values in the middle of the classes

- ▶ $(\text{Upper} + \text{lower limit}) / 2$

Class
Midpoints

64.5

74.5

84.5

94.5

104.5

114.5

124.5

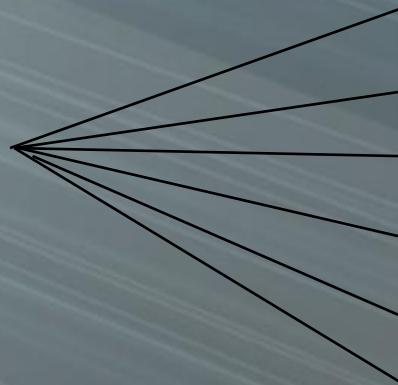
Table 2-2 Pulse Rates
of Females

Pulse Rate	Frequency
60-69	12
70-79	14
80-89	11
90-99	1
100-109	1
110-119	0
120-129	1

Class Width

is the difference between two consecutive lower class limits or two consecutive lower class boundaries

**Class
Width**



10

10

10

10

10

10

Table 2-2 Pulse Rates of Females

Pulse Rate	Frequency
60-69	12
70-79	14
80-89	11
90-99	1
100-109	1
110-119	0
120-129	1

Collect data

Bill
42.19
38.45
29.23
89.35
118.04
110.46
0.00
72.88
83.05
.
.

(There are 200 data points)

Prepare a frequency distribution

How many classes to use?

# of observations	# of classes
Less than 50	5-7
50 - 200	7-9
200 - 500	9-10
00 - 1,000	10-11

Class width = [Range] / [# of classes]
[119.63 - 0] / [8] = 14.95  15

Largest observation: 119.63

Smallest observation: 0

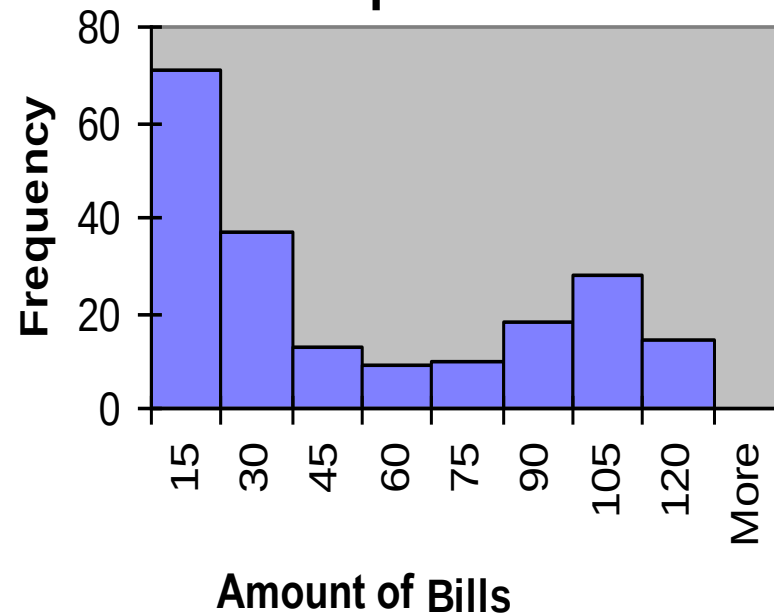
Range = Largest observation – Smallest observation

Frequency distribution of 200 long-distance telephone bills

Class Limits	Frequency
0 but less than or equal to 15	71
15 but less than or equal to 30	37
30 but less than or equal to 45	13
45 but less than or equal to 60	9
60 but less than or equal to 75	10
75 but less than or equal to 90	18
90 but less than or equal to 105	28
105 but less than or equal to 120	14
Total	200

<i>Bill</i>	<i>Frequency</i>
15	71
30	37
45	13
60	9
75	10
90	18
105	28
120	14
More	0

Histogram of 200 long-distance telephone bills



What information can we extract from this histogram

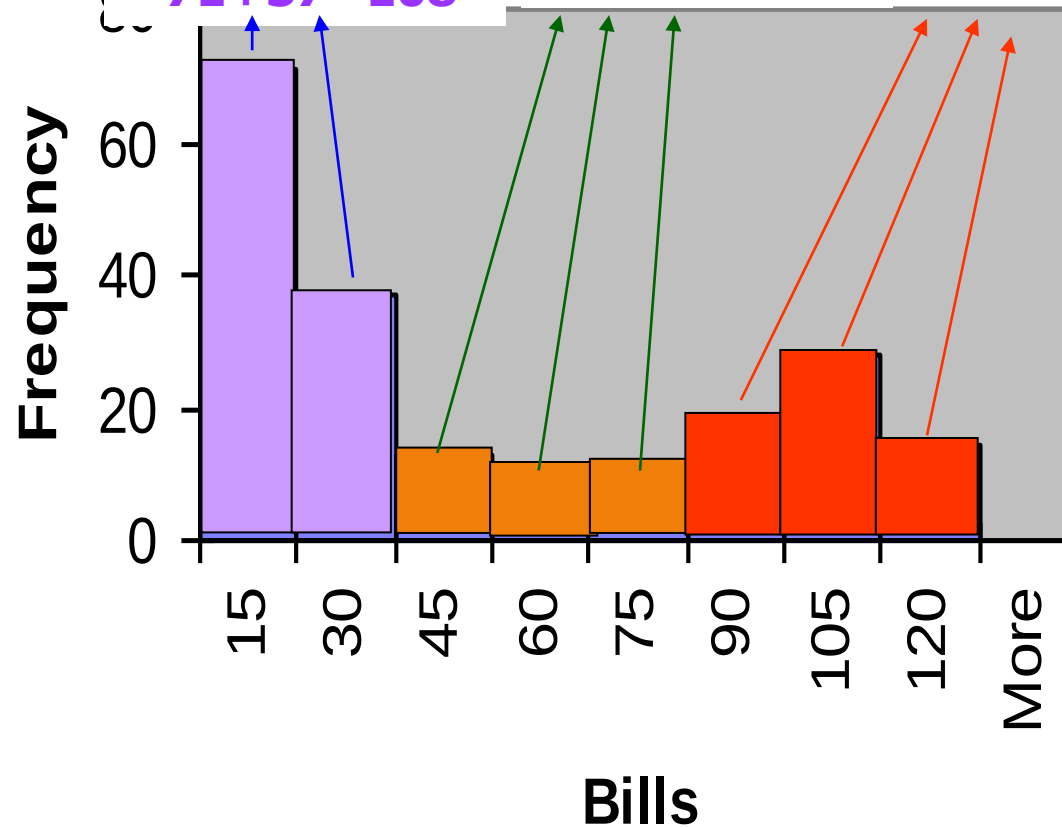
About half of all
the bills are small

$$71 + 37 = 108$$

A few bills are in
the middle range

$$13 + 9 + 10 = 32$$

Relatively, large
number of large
bills $18 + 28 + 14 = 60$



Relative Frequency

- ▶ It is sometimes preferable to show the relative frequency (proportion) of observations falling into each class, rather than the frequency itself.

$$\text{Relative frequency} = \frac{\text{Class frequency}}{\text{Total number of observations}}$$

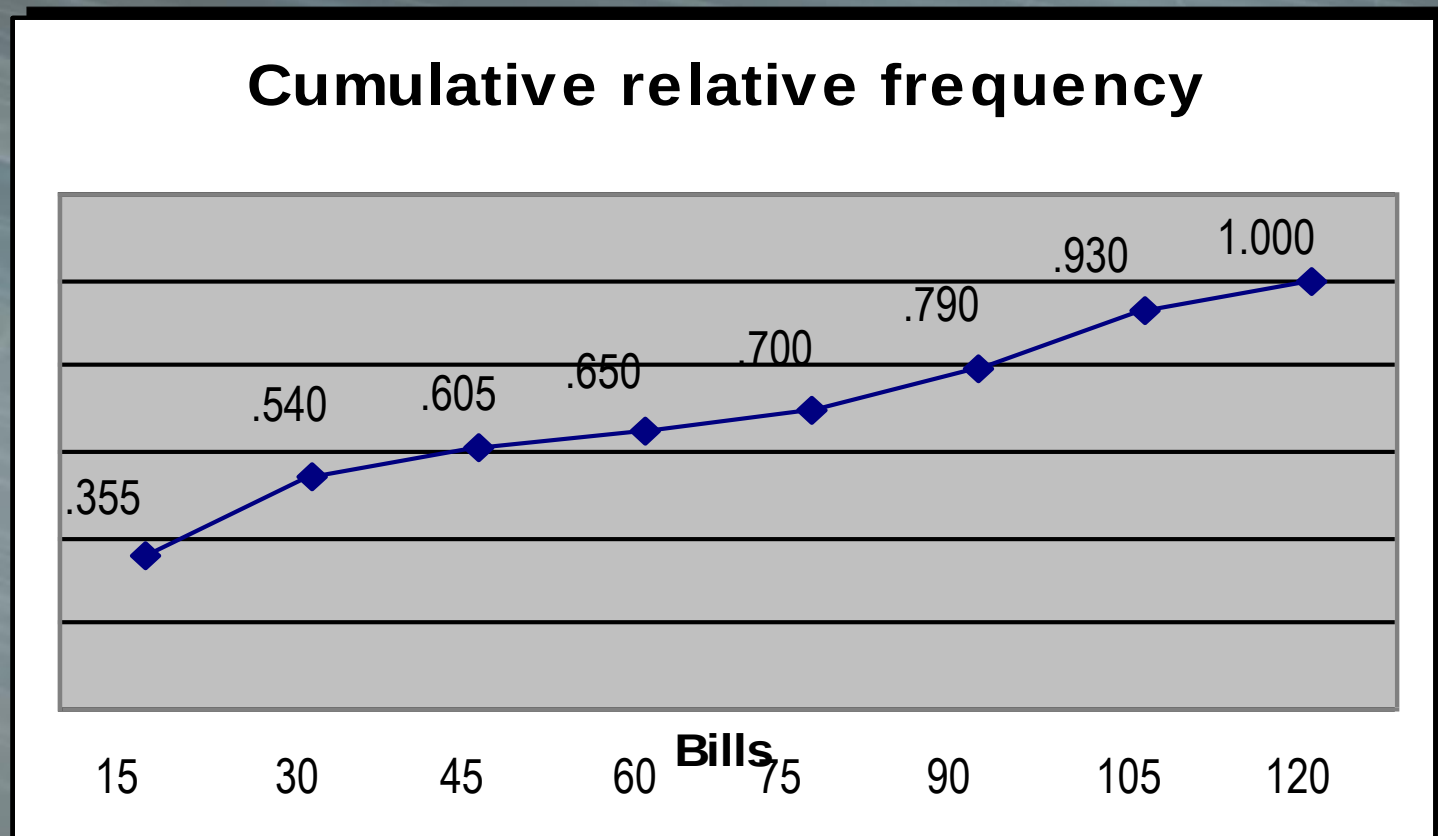
- ▶ Relative frequency is useful in comparing distributions while the number of observations of them are different.

Cumulative Relative Frequency Distributions

Cumulative relative frequency for telephone bills

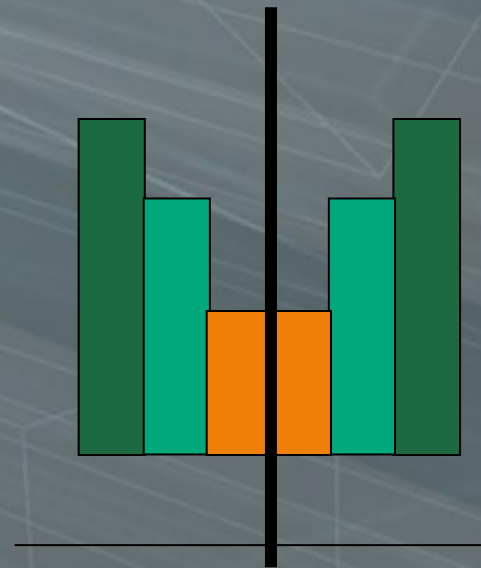
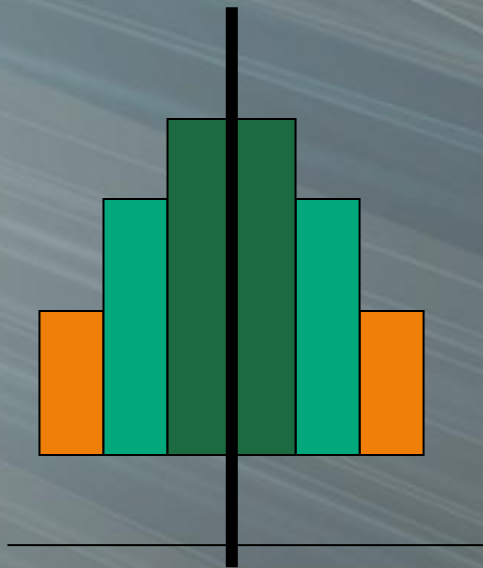
Class	Frequency	Cumulative frequency	Cum.Relative frequency
0-15	71	71	$71/200=.355$
15-30	37	108	$108/200=.540$
30-45	13	121	$121/200=.605$
45-60	9	130	$130/200=.650$
60-75	10	140	$140/200=.700$
75-90	18	158	$158/200=.790$
90-105	28	186	$186/200=.930$
105-200	14	200	$200/200=1.000$

Cumulative Relative Frequency Distributions



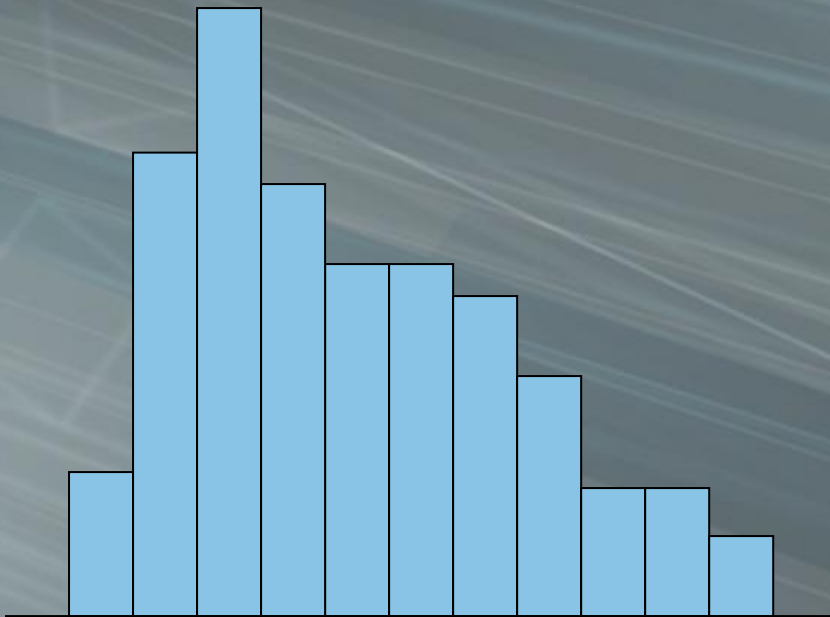
Shapes of Distribution

► Symmetry

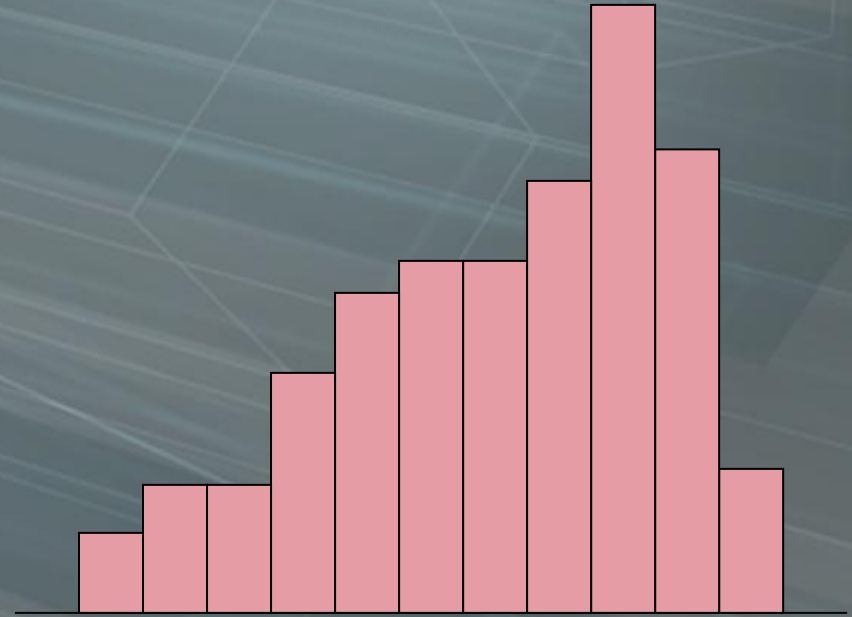


Shapes of Distribution

► Skewness



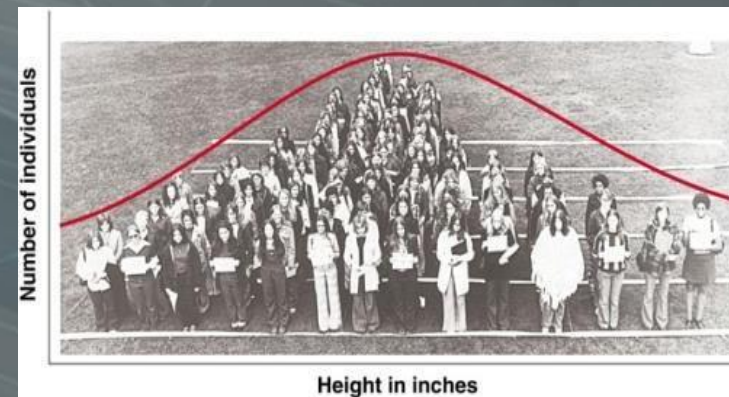
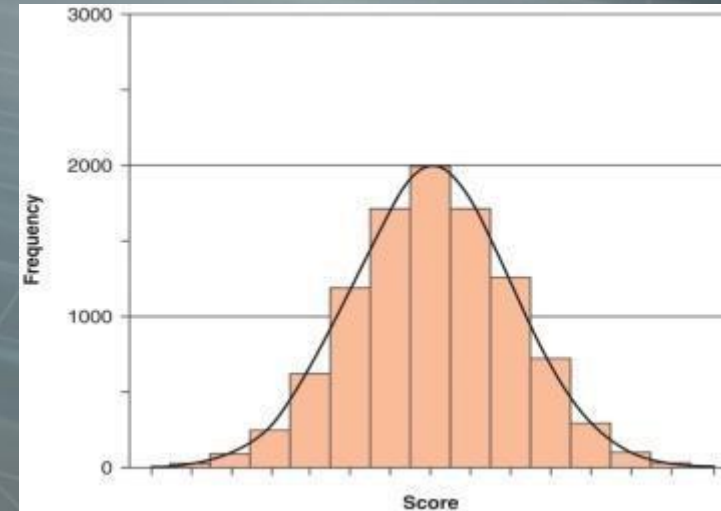
Positively skewed
(skewed to the right)



Negatively skewed
(skewed to the left)

Normal Distribution

- ▶ The properties of the normal distribution include:
 - ▶ bell shaped, symmetry around the middle point (unimodal), scores decline smoothly (short-tails) as we move away from middle point
- ▶ The mean, mode, median should all have similar values



Normal Distribution

- ▶ Having a well defined characteristics has allowed mathematicians/statisticians to develop tables of results which we can apply to solve problems.
- ▶ For many normal like distributions one standard deviation (SD) either side of the mean always covers 68.27% of the data.
- ▶ For two SD's it covers 95.44% of the data
- ▶ This is the basis on which parametric statistical tests such as t-test, anova and linear correlation operate

Tables

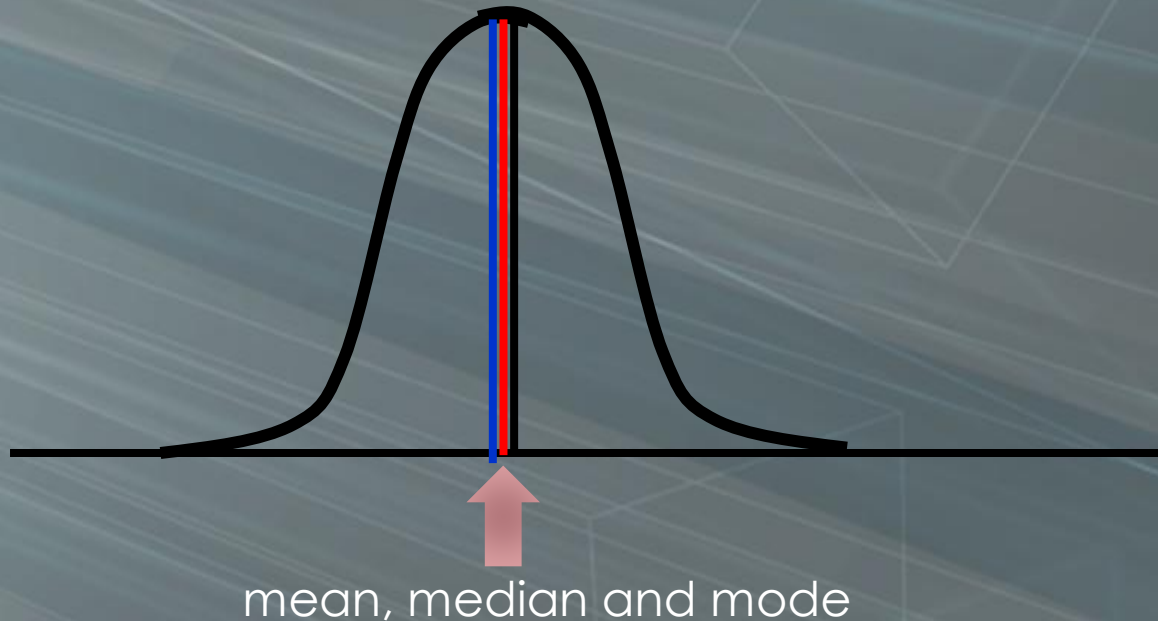


A NORMAL TABLE

z	Height	Area	z	Height	Area	z	Height	Area
0.00	39.89	0	1.50	12.05	89.64	3.00	0.443	99.730
0.05	39.84	3.99	1.55	11.90	87.69	3.05	0.487	99.771
0.10	39.69	7.97	1.60	11.75	85.64	3.10	0.527	99.806
0.15	39.45	11.92	1.65	11.59	83.51	3.15	0.570	99.837
0.20	39.10	15.85	1.70	11.43	81.06	3.20	0.615	99.864
0.25	38.67	19.74	1.75	11.27	78.09	3.25	0.663	99.885
0.30	38.14	23.58	1.80	11.10	74.81	3.30	0.712	99.905
0.35	37.52	27.37	1.85	10.93	71.07	3.35	0.762	99.920
0.40	36.82	31.08	1.90	10.75	66.80	3.40	0.813	99.933
0.45	36.05	34.73	1.95	10.56	62.00	3.45	0.865	99.944
0.50	35.21	38.29	2.00	10.36	56.66	3.50	0.917	99.953
0.55	34.29	41.77	2.05	10.15	50.78	3.55	0.970	99.960
0.60	33.29	45.12	2.10	9.93	44.29	3.60	0.983	99.968
0.65	32.22	48.33	2.15	9.69	37.29	3.65	0.994	99.974
0.70	31.07	51.41	2.20	9.44	29.67	3.70	0.999	99.979
0.75	29.84	54.37	2.25	9.18	21.48	3.75	0.999	99.982
0.80	28.53	57.20	2.30	8.91	12.80	3.80	0.999	99.984
0.85	27.14	60.00	2.35	8.63	3.59	3.85	0.999	99.986
0.90	25.67	62.67	2.40	8.34	-2.24	3.90	0.999	99.988
0.95	24.13	65.22	2.45	8.04	-8.97	3.95	0.999	99.990
1.00	22.50	67.67	2.50	7.73	-16.76	4.00	0.999	99.993
1.05	20.80	70.00	2.55	7.41	-24.92	4.05	0.999	99.995
1.10	19.05	72.24	2.60	7.08	-33.07	4.10	0.999	99.997
1.15	17.25	74.39	2.65	6.74	-41.20	4.15	0.999	99.998
1.20	15.40	76.43	2.70	6.39	-49.31	4.20	0.999	99.999
1.25	13.50	78.37	2.75	6.03	-57.40	4.25	0.999	99.999
1.30	11.56	80.20	2.80	5.66	-65.49	4.30	0.999	99.999
1.35	9.58	81.93	2.85	5.28	-73.56	4.35	0.999	99.999
1.40	7.57	83.55	2.90	4.89	-81.61	4.40	0.999	99.999
1.45	5.54	85.07	2.95	4.49	-89.64	4.45	0.999	99.999

Normal Distribution and Measure of Center

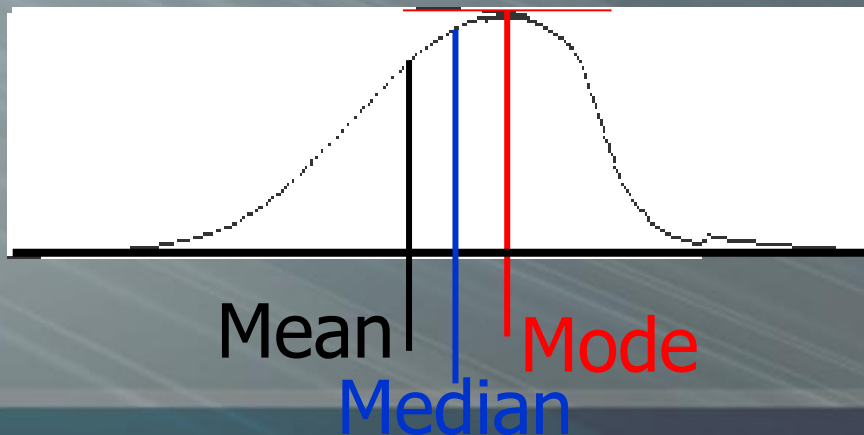
- ▶ In a normal distribution, the mean, median and mode should coincide.



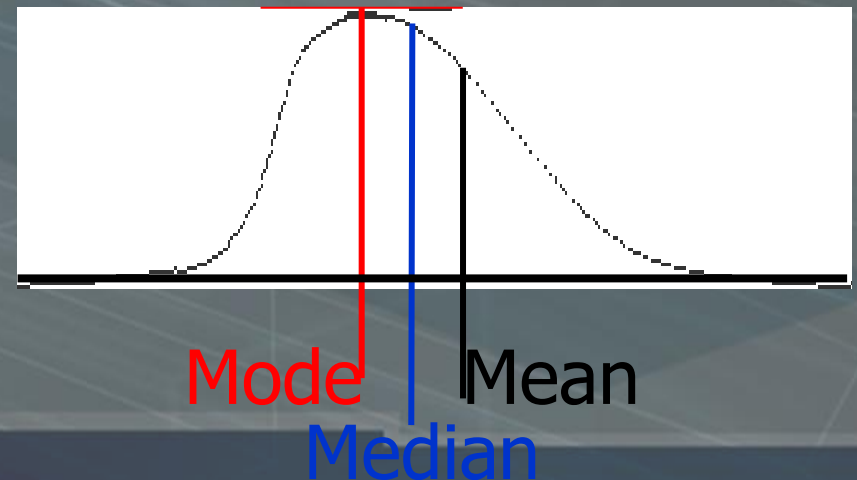
Normal Distribution and Measure of Center

- ▶ If distribution is skewed to the left or to the right, the three measures differ

A negatively skewed distribution
("skewed to the left")



A positively skewed distribution
("skewed to the right")



Question

- ▶ An automobile manufacturer carried out automobile mileage and gasoline-consumption testing. Ten models of automobile were road tested for 200 km in both city and highway driving conditions. The following data were recorded for km-per-litre performance.

City: 21.7 20.7 18.7 17.2 19.9 21.8 20.8 20.9
19.9 19.8

Highway: 26.8 23.8 24.2 25.0 22.6 22.3 24.2
24.7 27.4 25.2

- ▶ Use the mean, median and mode to compare the performance for city and highway driving.

Solution

► City

Mean = $(21.7 + 20.7 + 18.7 + 17.2 + 19.9 + 21.8 + 20.8 + 20.9 + 19.9 + 19.8)/10 = 20.14$

Median = $(19.9 + 20.7)/2 = 20.3$; Mode = 19.9

► Highway

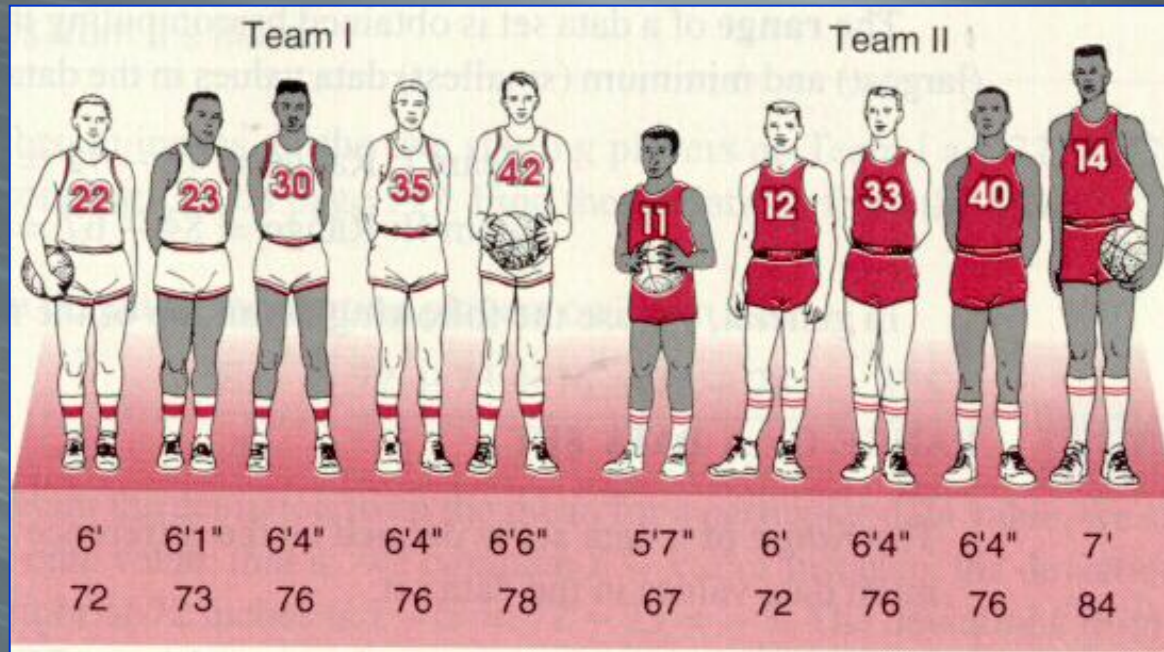
Mean = $(26.8 + 23.8 + 24.2 + 25.0 + 22.6 + 22.3 + 24.2 + 24.7 + 27.4 + 25.2)/10 = 24.62$

Median = $(24.2 + 24.7)/2 = 24.45$; Mode = 24.2

From the samples we can see that the mean, median and modal mileages are all better in the highway than in the city.

Dispersion

- ▶ It is clear that the heights of the two teams differ.
- ▶ The heights of the players on Team II vary much more than those on Team I



Same mean (75 inches)
Same mode (76 inches)
Same median (76 inches)

Measures of variability

- ▶ Measures of central location fail to tell the whole story about the distribution of a given data set.
- ▶ We need to measure the varies around the mean value
 - ▶ Variance
 - ▶ Standard Deviation

Sample Variance

- ▶ The variance of a sample of n measurements $x_1, x_2, \dots, x_n$ having a mean $\bar{x}$ is defined as

Sample Value – Sample Mean

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1}$$

Example

- Find the mean and the variance of the following sample of measurements (in years).

3.4, 2.5, 4.1, 1.2, 2.8, 3.7

$$\bar{x} = \frac{\sum_{i=1}^6 x_i}{6} = \frac{3.4 + 2.5 + 4.1 + 1.2 + 2.8 + 3.7}{6} = \frac{17.7}{6} = 2.95$$

$$\begin{aligned} s^2 &= \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n-1} \\ &= \frac{(3.4 - 2.95)^2 + (2.5 - 2.95)^2 + \dots + (3.7 - 2.95)^2}{6-1} = 1.075 \text{ years}^2 \end{aligned}$$

Standard Deviation

- ▶ The standard deviation of a set of measurements is the square root of the variance of the measurements.

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

Question

- ▶ Rates of return over the past 10 years for two mutual funds are shown below. Which one have a higher level of risk?
- ▶ Fund A: 8.3, -6.2, 20.9, -2.7, 33.6, 42.9, 24.4, 5.2, 3.1, 30.05
- ▶ Fund B: 12.1, -2.8, 6.4, 12.2, 27.8, 25.3, 18.2, 10.7, -1.3, 11.4

► Solution

Refer to the Excel printout that is run from the “Descriptive statistics” sub-menu

Fund A should be considered more risky because its standard deviation is larger

<i>Fund A</i>		<i>Fund B</i>	
Mean	16	Mean	12
Standard Error	5.295	Standard Error	3.152
Median	14.6	Median	11.75
Mode	#N/A	Mode	#N/A
Standard Deviation	16.74	Standard Deviation	9.969
Sample Variance	280.3	Sample Variance	99.37
Kurtosis	-1.34	Kurtosis	-0.46
Skewness	0.217	Skewness	0.107
Range	49.1	Range	30.6
Minimum	-6.2	Minimum	-2.8
Maximum	42.9	Maximum	27.8
Sum	160	Sum	120
Count	10	Count	10

The Range

- ▶ The range of a set of measurements is the difference between the largest and the smallest measurements.
- ▶ Its major advantage is the ease with which it can be computed.
- ▶ Its major shortcoming is its failure to provide information on the dispersion of the values between the two end points.

Quartiles

- ▶ Quartiles
 - ▶ $Q1$ = lower quartile = $(n+1)/4$ ordered observation
 - ▶ $Q2$ = median = $(n+1)/2$ ordered observation
 - ▶ $Q3$ = upper quartile = $3(n+1)/4$ ordered observation
- ▶ Inter-quartile range (IQR四分位距)
 $IQR = Q3 - Q1$

Example

- ▶ Find the quartiles of the following set of measurements

7, 8, 12, 17, 29, 18, 4, 27, 30, 2, 4, 10, 21, 5, 18

- ▶ First sort the 15 measurements:

2, 4, 4, 5, 7, 8, 10, 12, 17, 18, 18, 21, 27, 29, 30

Q1 is at $(15+1)/4 = 4$

The 1st quartile

At most $(0.25)(15) = 3.75$ measurements should appear below the first quartile.

At most $(0.75)(15) = 11.25$ measurements should appear above the first quartile.

So, the lower quartile or the first quarter Q_1 is 5.

Similarly, the second quartile Q_2 is 12, which is the median.

The third quartile or the upper quartile Q_3 is 21.

Question

- ▶ The records of a company show the following numbers of days of sick leave taken by its employees for a sample of days during the summer months and a sample of days during the winter months.

Summer	Winter
18	28
20	18
15	24
16	32
21	18
20	29
12	23
16	38
19	28
20	18

- (a) Compute the range and the inter-quartile range for each period.
- (b) Compute the mean, variance and standard deviation for each period.

Solution

► (a) Summer: **Range** = $21 - 12 = 9$

$$Q_3 = \frac{3(n+1)}{4} = 8.25^{\text{th}} = \text{The average of } 8^{\text{th}} \text{ and } 9^{\text{th}} \text{ items,}$$

$$\text{i.e. } Q_3 = \frac{20 + 20}{2} = 20$$

$$Q_1 = \frac{(n+1)}{4} = 2.75^{\text{th}} = \text{The average of } 2^{\text{nd}} \text{ and } 3^{\text{rd}} \text{ items,}$$

$$\text{i.e. } Q_1 = \frac{15 + 16}{2} = 15.5$$

$$IQR = Q_3 - Q_1 = 20 - 15.5 = 4.5$$

Solution (cont')

► Winter:

$$\text{Range} = 38 - 18 = 20$$

$$IQR = Q_3 - Q_1 = 30.5 - 18 = 12.5$$

(b)

	Mean	Variance	Standard Dev.
Summer	17.7	8.2333	2.8694
Winter	25.6	44.4889	6.6700

Interval Estimate

Point Estimate and Interval Estimate

- ▶ **Point Estimate** (e.g. Mean, mode, median), gives the best guess from sample
- ▶ We also want to be able to say something that expresses the degree of certainty that we have in our guess
- ▶ An **Interval Estimate** gives a range of plausible values for a population parameter.

Margin of Error

- ▶ One common form of interval estimate:

statistic $\pm$ margin of error

- ▶ Where the margin of error reflects the precision of the sample statistics as a point estimate for the parameter
- ▶ E.g. mean is **44.6** with margin of error **± 2.23**

Examples

中大香港亞太研究所公布 2024 年 3 月份 特區政府及特首民望意見調查結果

附表二：對特首李家超整體表現的評分（平均分數）

	整體表現評分*	(樣本數)
24 年 3 月〔註一〕	44.6	(676)
23 年 9 月	47.9	(698)
23 年 7 月	50.4	(674)
23 年 5 月	51.0	(694)
23 年 3 月〔註二〕	50.8	(689)

問題：「整體嚟講，以 0 分為最低分至 100 分為最高分，50 分為合格，你會俾特首李家超整體表現幾多分呢？」

* 行政長官李家超 2024 年 3 月份評分的抽樣誤差（可信度設於 95%）為正負 2.23 分，標準差（standard deviation）為 29.596，中位數（median）為 50 分，眾值（mode）為 50 分。

註一：經 t 檢定顯示，2024 年 3 月和 2023 年 9 月的平均評分差異達統計上顯著水平 [$p < 0.05$]。

註二：經 t 檢定顯示，2024 年 3 月和 2023 年 3 月的平均評分差異達統計上顯著水平 [$p < 0.05$]。

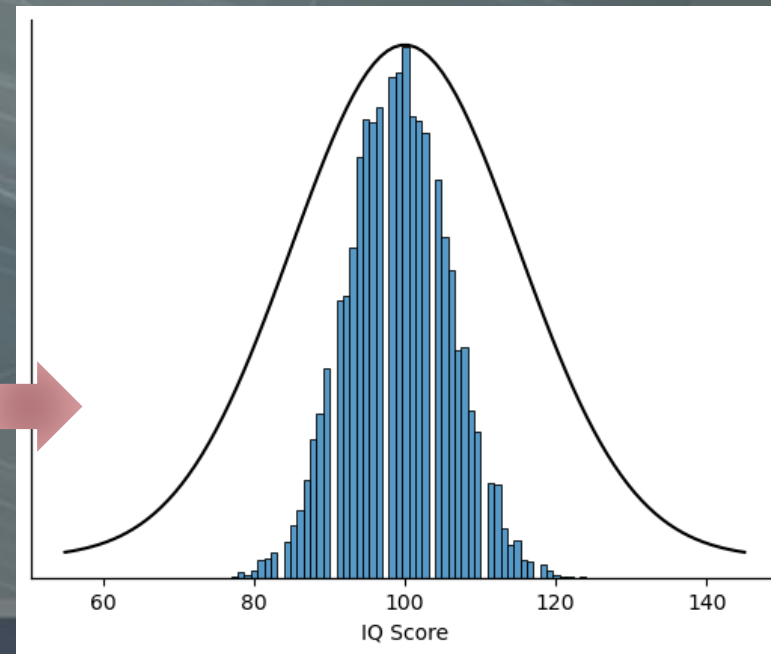
Margin of Error

- ▶ How do we determine the margin of error?
- ▶ We can use the spread of the sampling distribution (the standard error) to determine the margin of error.
- ▶ But what is sampling distribution?
 - ▶ To simplify the discussion afterwards, we assume our interested statistics is mean

Sampling Distribution

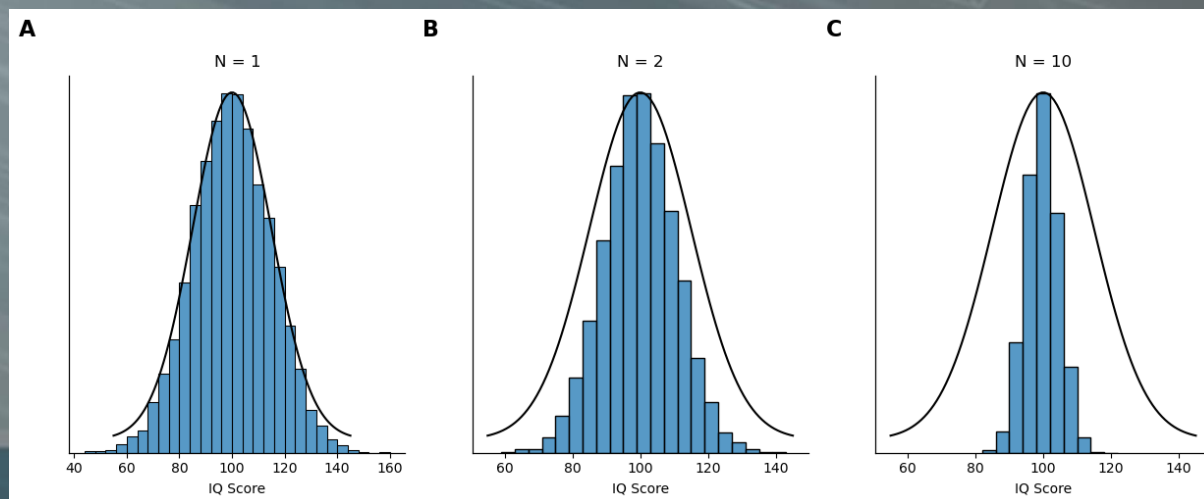
- ▶ When we repeatedly sample and compute the mean, e.g. 5 people each time for their IQ score
- ▶ We obtain a distribution called “**sampling distribution**” of the mean

	Person 1	Person 2	Person 3	Person 4	Person 5	Sample Mean
Replication 1	99	84	105	102	91	96.2
Replication 2	127	127	92	88	121	111.0
Replication 3	90	114	125	98	99	105.2
Replication 4	115	114	72	122	103	105.2
Replication 5	92	93	98	109	122	102.8
Replication 6	109	92	104	79	105	97.8
Replication 7	91	91	104	61	98	89.0
Replication 8	82	83	73	108	92	87.6
Replication 9	107	84	102	118	95	101.2
Replication 10	77	91	113	108	97	97.2



Properties of Sampling Distribution

- ▶ If the population is normal distributed, then the sampling distribution of mean will also be normal distributed
 - ▶ As sample mean should most likely to hit the population mean, and then spread around it
- ▶ The bigger the sample size, the narrower the sampling distribution



Central Limit Theorem

- ▶ The mean of the sampling distribution is the same as the mean of the population
- ▶ The standard deviation of the sampling distribution (i.e., the standard error) gets smaller as the sample size increases
- ▶ The shape of the sampling distribution becomes normal as the sample size increases
 - ▶ Even the population is not normal, the shape of sampling distribution will also be normal for large sampling size!

Central Limit Theorem

- ▶ The 2nd statement leads to the following formula :

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

- ▶ SEM stands for Standard Error of Mean
- ▶ σ is the population standard deviation
- ▶ N is the sampling size

Confidence Interval

- ▶ Back to our previous problem to find the Margin of error
- ▶ With the IQ score example, it would be nice to be able to say that there is a 95% chance that the true mean lies between 109 and 121. [NOTE: this is the claim we want, but we will find it inaccurate to describe it in this way]
- ▶ The name for this is a **confidence interval for the mean**.

Confidence Interval

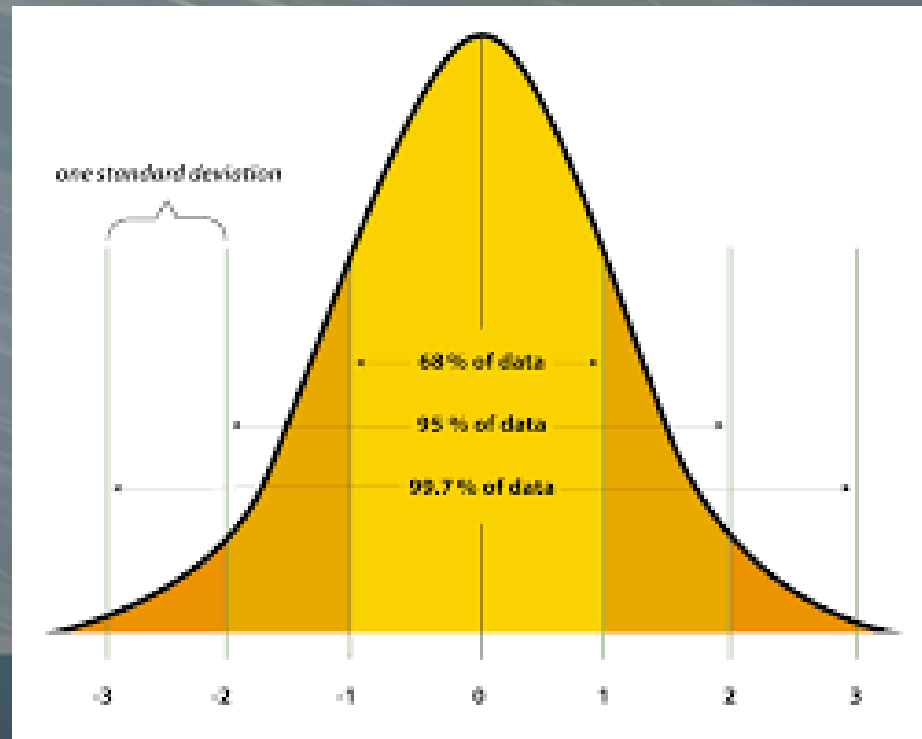
- ▶ If the sampling distribution is relatively symmetric and bell-shaped, a 95% confidence interval can be estimated using
 $\text{statistic} \pm 2 \times (\text{standard error})$

- ▶ So if our interested statistics is mean, it becomes:

$$\text{mean} \pm 2 \times \text{SEM}$$

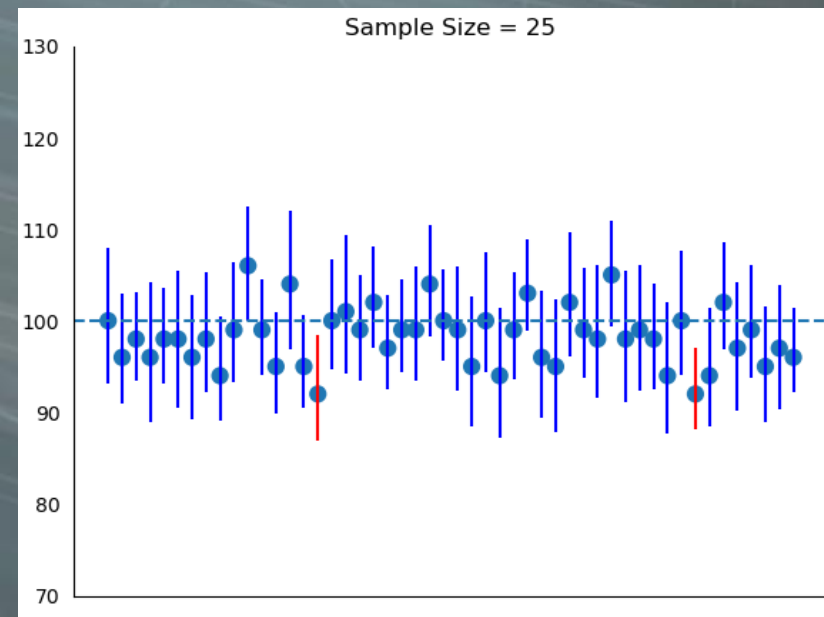
Confidence Interval

- ▶ We must be careful of the magic number “2” in the formulation
 - ▶ 2 standard deviations covers 95% of data



Confidence Interval

- ▶ The parameter is fixed
 - ▶ E.g. population mean
- ▶ The statistic is random (depends on the sample)
- ▶ The interval is random (depends on the statistic)
- ▶ 95% of 95% confidence intervals will capture the truth

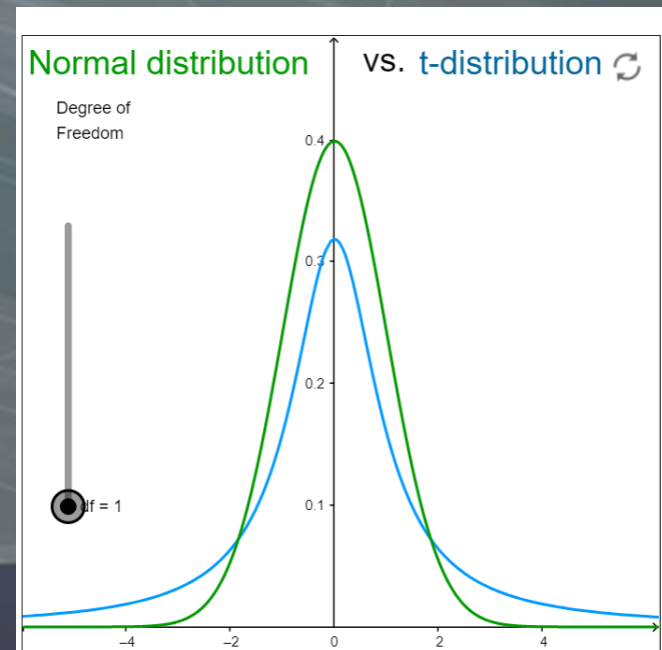


Confidence Interval

- ▶ One thing we missed, what is the population std. dev. ?

$$\text{SEM} = \frac{\sigma}{\sqrt{N}}$$

- ▶ We only estimate with sample std. dev.
- ▶ Usually t-distribution will be used to estimate the margin of error when N is small

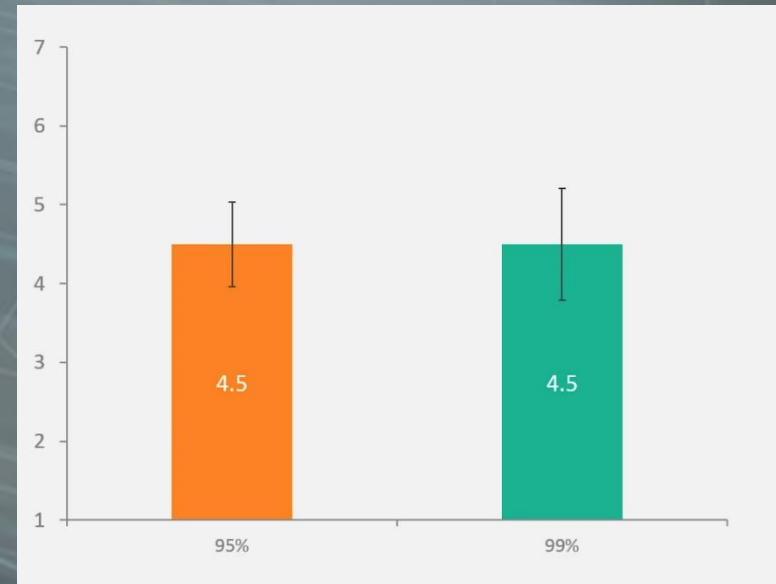


Confidence Interval

- ▶ When N is small, we get a much bigger number when we use the t distribution
- ▶ Bigger values mean that the confidence interval is wider, indicating that we're more uncertain about what the true value of actually is.

The confidence level

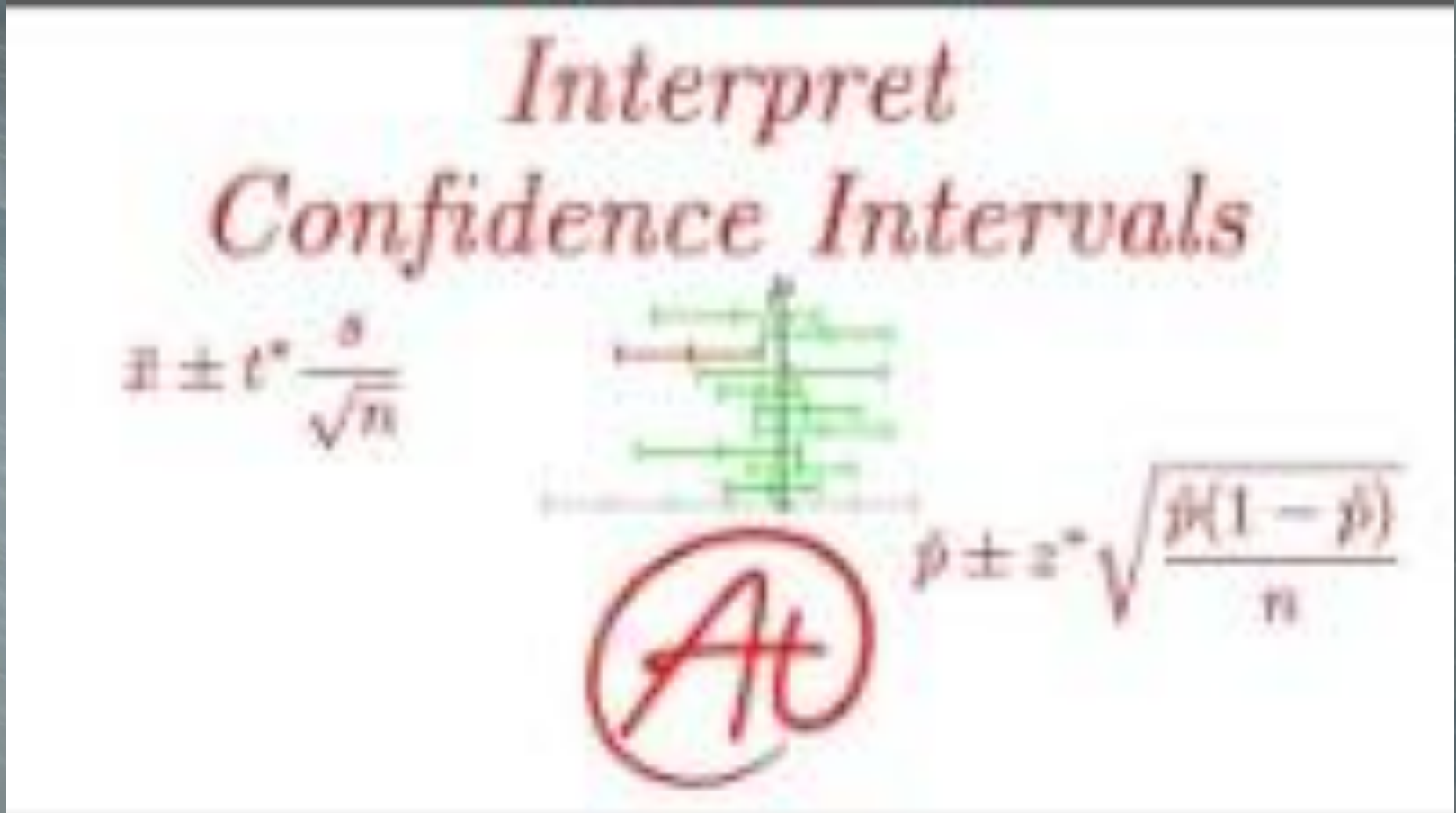
- ▶ 95% confidence level is the standard
- ▶ Other levels can also be used, e.g. 99% confidence intervals
 - ▶ A larger margin of error if other factors the same
- ▶ If we increase our sample size (and kept everything else the same) we will see our confidence intervals reduce.



Interpretation of Confidence Interval

- ▶ Care must be paid to the interpretation / claims of CI !
- ▶ As CI is also **random**, we have limitation to claim that we have 95% chance that the true mean falls in our interval
- ▶ 95% of all samples yield intervals that contain the true parameter, so we are 95% confident that *this interval* captures the true population mean

Interpretation of Confidence Interval



Common Misinterpretation

- ▶ Misinterpretation 1: “A 95% confidence interval contains 95% of the data in the population”
- ▶ Misinterpretation 2: “I am 95% sure that the mean of a sample will fall within a 95% confidence interval for the mean”
- ▶ Misinterpretation 3: “The probability that the population parameter is in this particular 95% confidence interval is 0.95”

Summary

- ▶ Data preparation process
- ▶ Data collection and sampling
- ▶ 4 Major Types of Data
 - ▶ Nominal, Ordinal, Interval, Ratio
- ▶ Encoding approaches
 - ▶ Ordinal and One-hot encoding
- ▶ Basic descriptive statistics
 - ▶ Central tendency
 - ▶ Mean, Median and Mode
 - ▶ Dispersion
 - ▶ Variance and Standard Deviation
 - ▶ Interval Estimate